DAY-1
FINAL AID
FMGE RAPID REVISION
Chanting-Style High-Yield Review
Anatomy
Physiology
Biochemistry
Pathology
Microbiology
Pharmacology
19
SUBJECTS
03
DAYS

DAY-2
FINAL AID
FMGE RAPID REVISION
Chanting-Style High-Yield Review
PSM
Forensic Medicine
ENT
Ophthalmology
Orthopaedics
Anaesthesia
19
SUBJECTS
03
DAYS

DAY-3
FINAL AID
FMGE RAPID REVISION
Chanting-Style High-Yield Review
Internal Medicine
Surgery
Obstetrics & Gynaecology
Paediatrics
Radiology
Dermatology
19
SUBJECTS
03
DAYS

FOR PDF = TELEGRAM = CONCEPTUAL MEDICINE
FOR DOUBT = INSTA = PJAMWAL7

RADIOLOGY

♦ **Discoveries =**

- → X-Ray - W.C. Roentgen
- → CT - Godfrey Hounsfield
- → MRI - Laterbeur & Mansfield
- → Radioactivity - Bacquerel

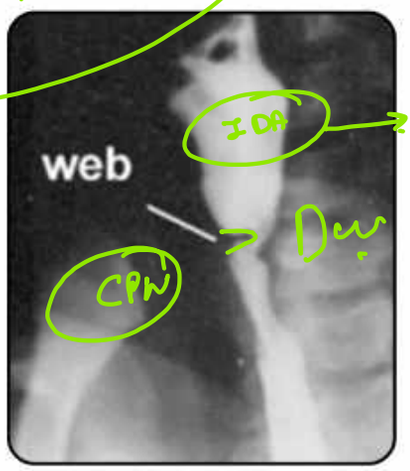
♦ **GIT =**

→ **Important contrast =**

1. Barium swallow → Esophagus
2. Barium meal → Stomach & Proximal Duodenum
3. Barium meal follow through → Small Bowl
4. Barium enema → Large Bowl

→ **Disease =**

1. Plummer Vinson Syndrome =
   - → Cricopharyngeal web
   - → Iron deficiency anaemia
   - → Dysphagia
   - → Glossitis



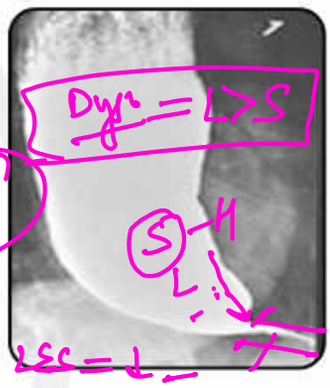
2. Zenker's Diverticulum

→ Outpouching of mucosa through weak muscle layer

3. Achalasia cardia

→ Due to failure of relaxation of lower Esophageal Sphincter

→ Bird beak appearance



4. Carcinoma Oesophagus

→ Rat tail appearance

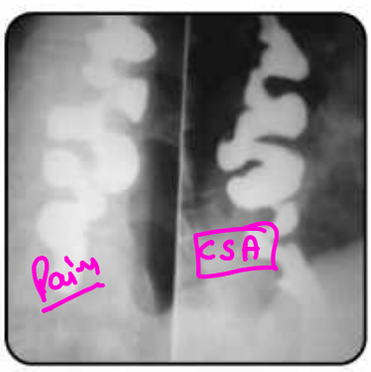
→ Upper 2/3rd Squamous cell carcinoma

→ Lower 1/3rd Adenocarcinoma

5. Diffuse Oesophageal Spasm

→ On X-Ray = Corkscrew appearance

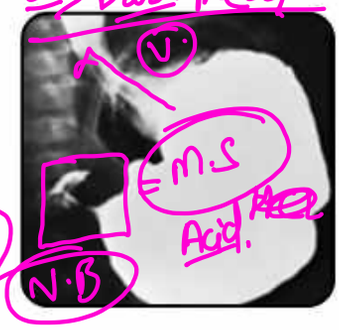
→ Painful condition due to abnormal contraction of oesophagus.



↑Size of Cell

6. Congenital Hypertrophic Pyloric Stenosis
   → Presents few weeks after birth
   → Child presents with non-bilious vomiting
   → Per abdomen = Olive shape lump
   ⇒ M.A

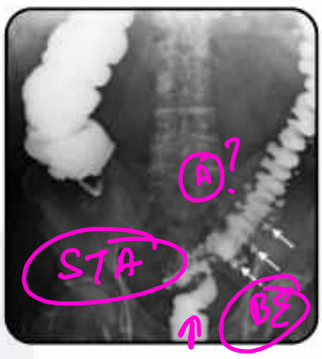
⇒ Barr meal



OSL

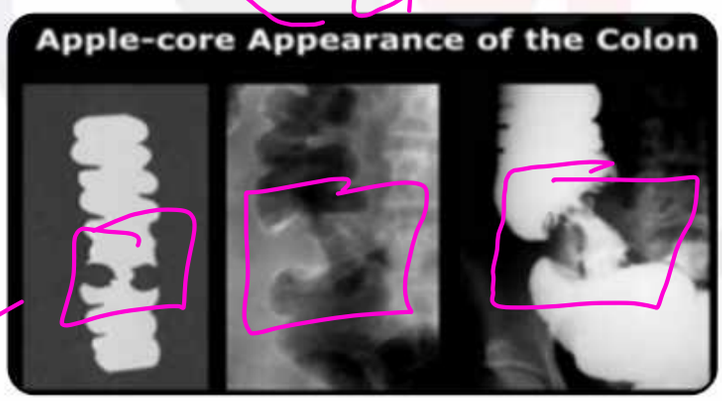
7. Diverticulitis =
   → Saw tooth appearance
   I.O.C. → CECT

30 → VE
40-50 → MG
20 → T

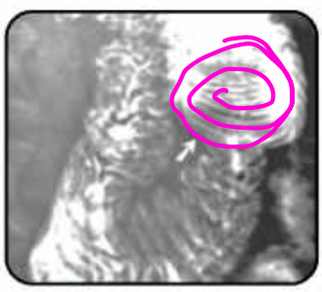


8. Colon Cancer = Rt / Lt

X-Ray Barr meal

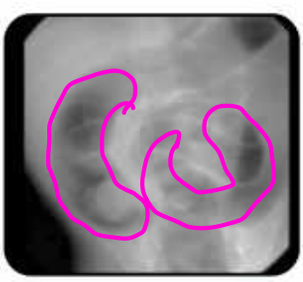


9. Intussusception + RRJS


Coiled spring app.

Claw sign

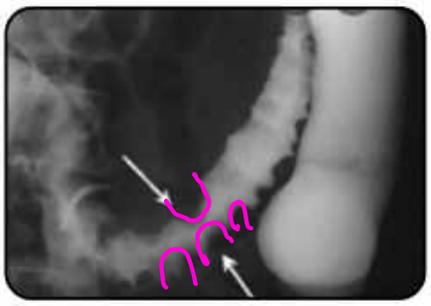
Doughnut Sign


Saucage shape lump

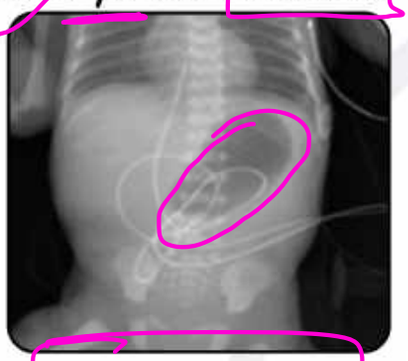
= Vaccine ? = ROTA virus Target Gp

10. Ischemic Colitis =

→ Thumb printing sign is seen

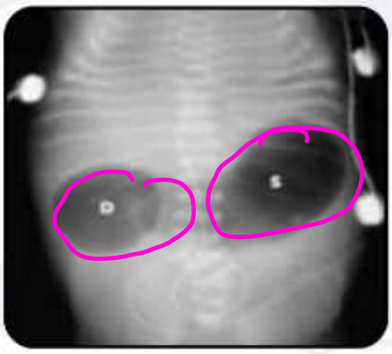


11. Pyloric Atresia



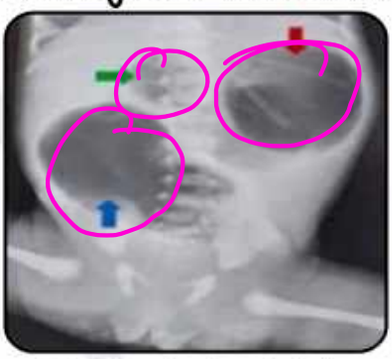
Single bubble

12. Duodenal Atresia



Double bubble

13. Jejunal Atresia

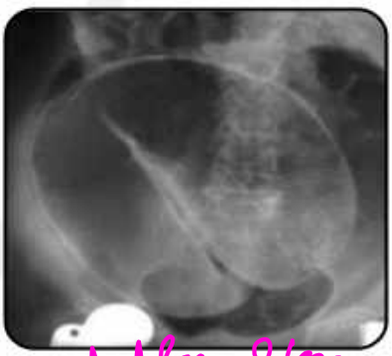


Tripple bubble

14. Sigmoid Volvulus

→ Sigmoid colon wrap against itself (anti-clock wise)

→ Coffee bean appearance seen



15. Ulcerative Colitis =

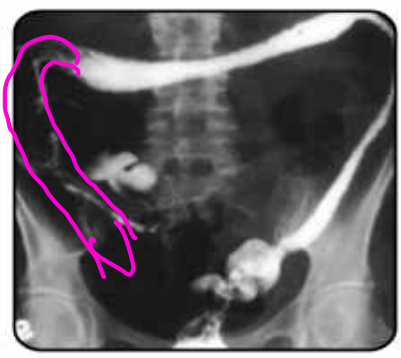
→ Lead pipe appearance

→ Collar button ulcer

→ Pseudo Polyp's seen

→ Granular mucosa seen

→ Submucosal or Mucosal involvement



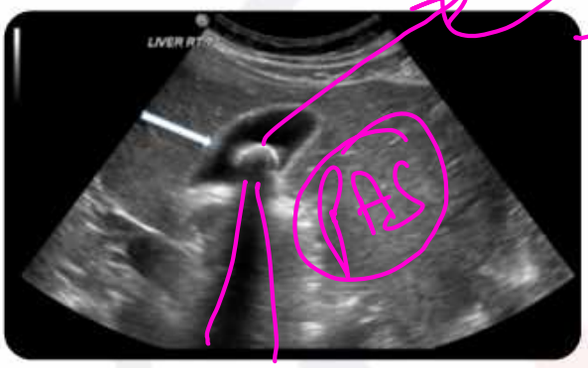
16. Hydatid Cyst =

→ Cause echinococcus / dog tape worm

→ On USG → Water Lilly sign

D = Dog
I = M/Ch

17. Gall Stone



→ ? USG

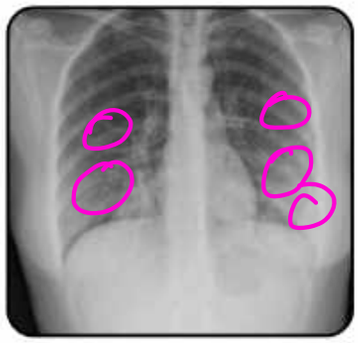
18. Congenital Diaphragmatic Hernia

Rt. Lung

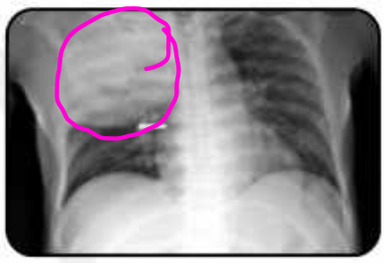
19. Crohn's Disease

Respiratory System

1. Atypical Pneumonia
   → Mcc = Mycoplasma



2. Klebsiella Pneumonia
   → Bulging fissure sign +nt

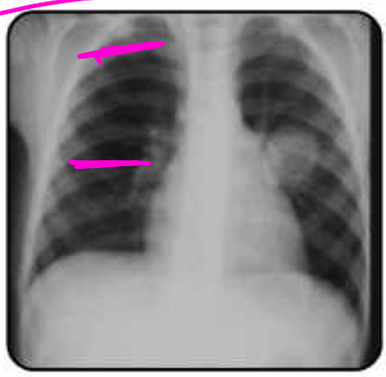


3. Aspergilloma
   → Fungal infection of lung
   caused by = Aspergillus

Air crescent - On CT scan

4. Miliary TB =
   → Hematogenous route spread
   → Millet seed like pattern is seen

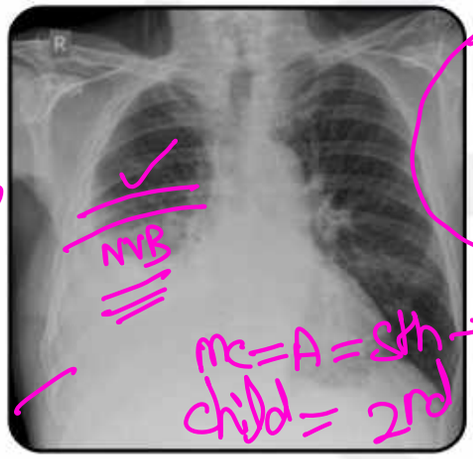
5. GHONS Focus = 1° = mc chd → mid L
   → Seen in primary TB
   → +nt in middle to lower lobe



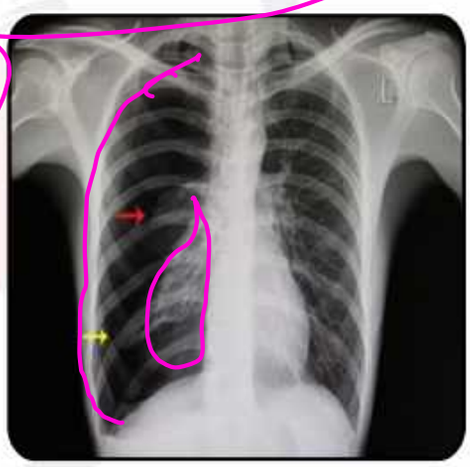
6. Simon Focus = Seen in secondary TB
   = Apical part or lung is involved

2° = m+A = Apical + Sec

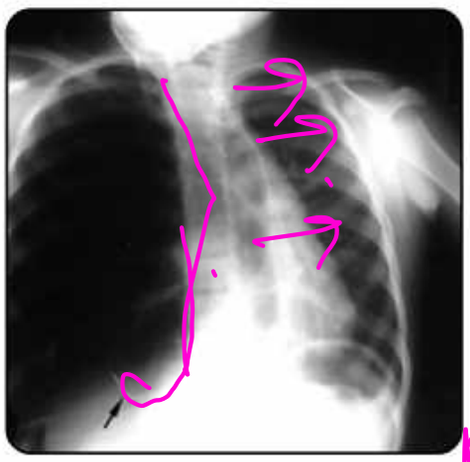
7. Pleural Effusion



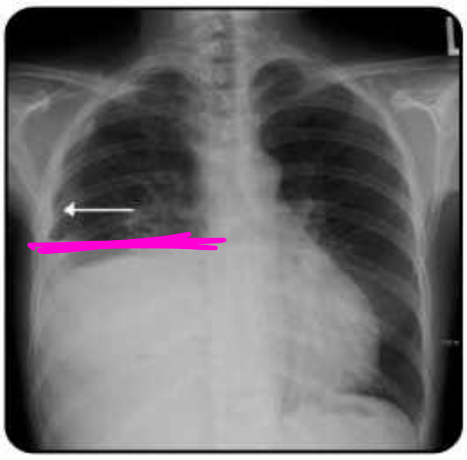
8. Pneumothorax



9. Tension Pneumothorax



10. Hydropneumothorax



11. Cannon Ball appearance

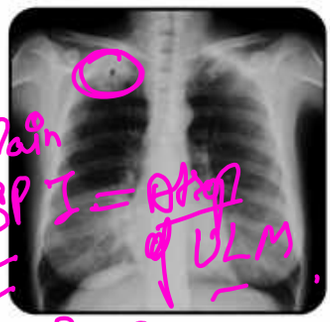
→ Seen in lung metastasis

12. Pancoast Tumour

→ Apical mass on RT side

→ It can give rise to

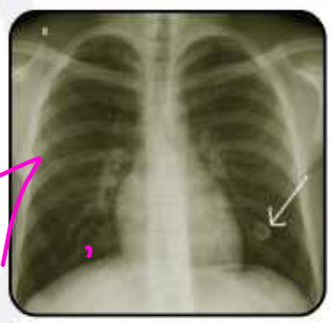
- Pancoast Syndrome
- Horner's Syndrome



13. Pulmonary Hamartoma

→ Popcorn calcification on X-Ray

(Mixture of tissue components)



14. Bronchiectasis =

→ Abronmal dilatation of bronchi

→ On CT scan - signet ring app. is seen

15. Sarcoidosis =

→ Abnormal growth of inf. cells (granuloma's) in lung

→ On X-Ray B/L hilar lymphadenopathy seen

16. Emphysema

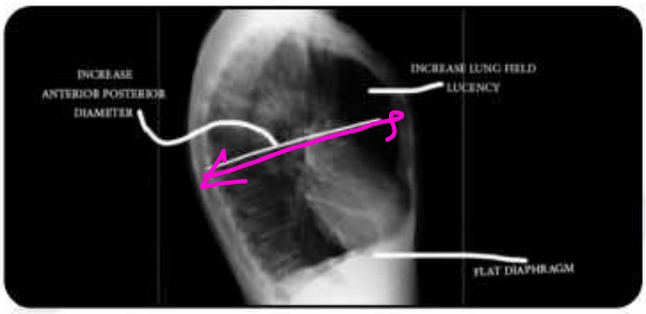
On X Ray barrel shape chest is seen

→ MC Cause - Smoking

OLD

CA = Smoking → SCC

PA = α1 ATD = Asy.



VC = ↑

FEV1% = ↓↓

17. Silicosis

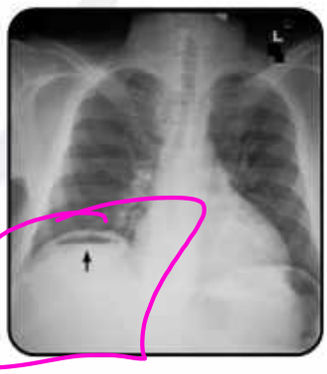
Egg Cell Calcification

18. Pneumo Peritonium

Next step: I.V. Fluid

↓

Exploratory Laprotomy



Cupula Sign

19. Consolidation with Air Bronchogram

↓

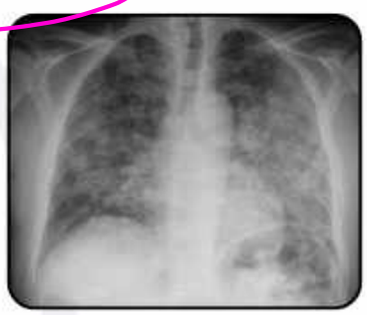
Air relapsed by Fluid

20. COVID-19

Multifocal patchy ground glass opacity

21. Pnuemocystis Jiroveci

C/P: HIV positive + B/L Ground glass appearance

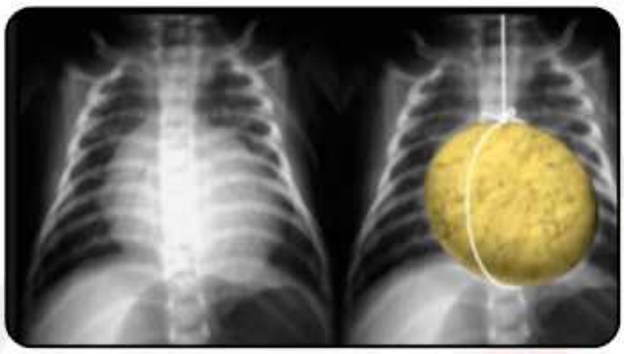


Cardio Vascular System

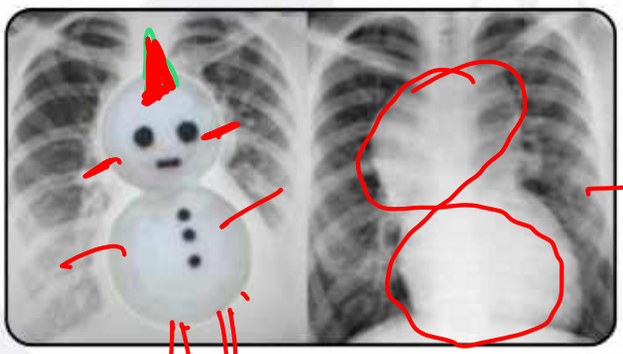
1. TGA - Transposition of great arteries

→ On X-Ray = Egg on string app. seen

→ Aorta abnormally arise from Right Ventricle



2. TAPVR = Total Anomalous Pulmonary Venous Return

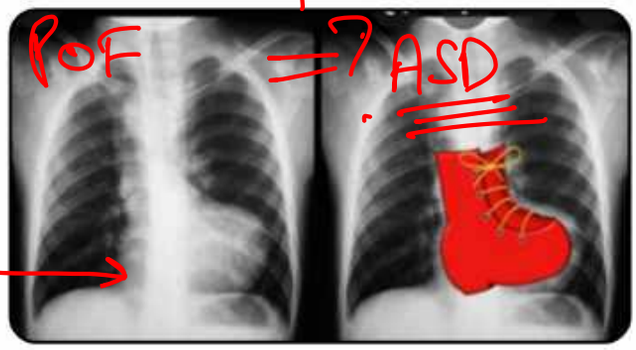


→ On X-Ray figure of eight appearance

→ Pulmonary vein drain into right atria

3. TOF = Tetralogy of Fallot

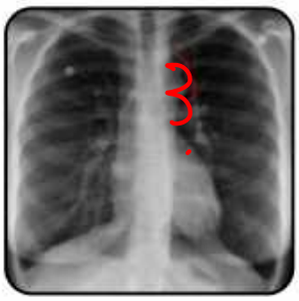
- 4 Components →
  - → Pulmonary Stenosis
  - → VSD
  - → Overriding Of Aorta
  - → RVH
- On X-Ray → Boot shaped heart seen



4. Coartcation of Aorta

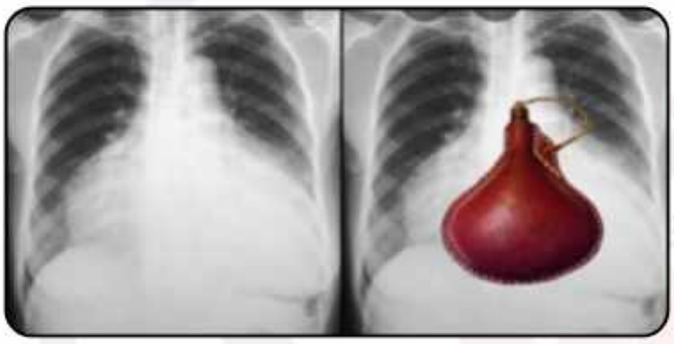
→ Figure of 3 app. seen

→ Persistent narrowing of Aorta



5. Pericardial Effusion

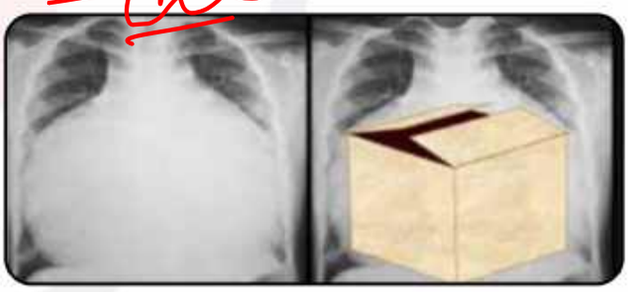
→ Money bag/ Leather bottle appearance



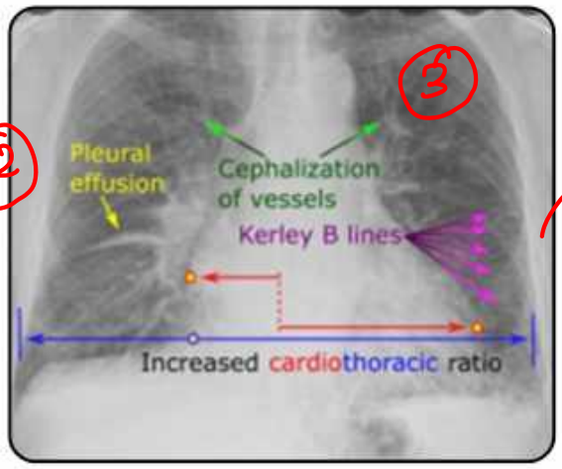
6. Ebstein Anomaly

→ Box-shaped heart

→ Abnormal enlarged ventricles



7. CHF = Congestive Heart Failure



8. Constrictive Pericarditis

→ Sign : Egg in cup sign

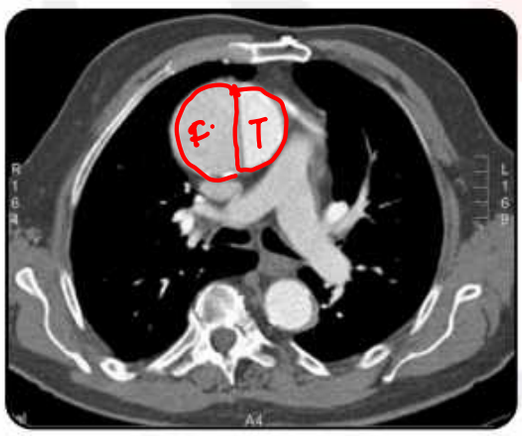
→ Thickening and calcification of Pericardium

→ MC Seen in TB

9. Aortic Dissection

→ A tear in Inner layer of Aorta will lead to Tennis Ball Appearance

[True Lumen Whiter than False]



8 → ? =

10F → ?

0 E in Cup

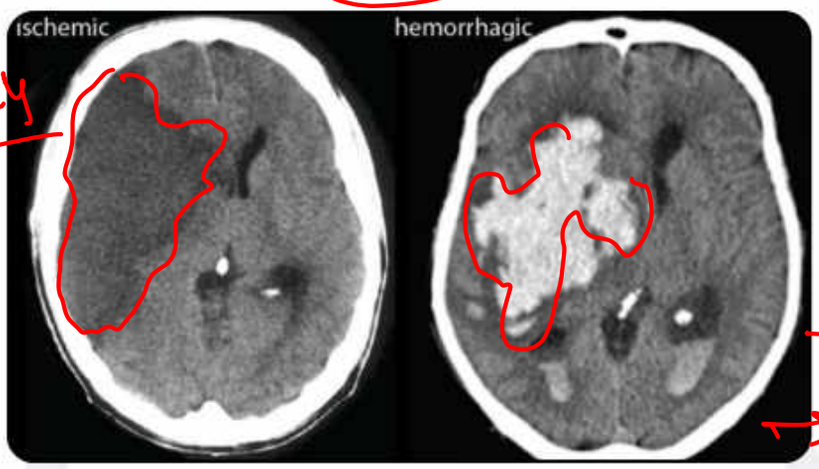
E on S

ESC

CNS

1. Ischemic Stroke

NCCT



→ Thromboembolism

Rx - Aspirin

2. Haemorrhagic Stroke

→ HTN
→ ED
→ Smoking

→ Area = Putamen

→ Rupture MCA

3. Subarachnoid Haemorrhage

→ Thunder clap headache

→ Bleeding in subarachnoid space

BA

4. Subdural Haemorrhage

Subka Paas = Bridging veins

→ Sickle shape haemorrhage

→ Injury to the bridging veins

Idli

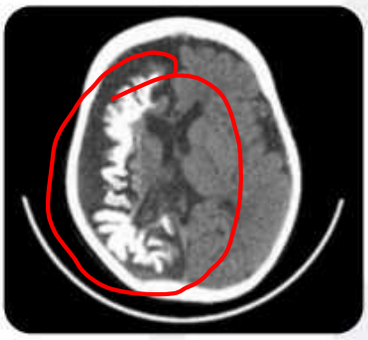
5. Extra Dural Haemorrhage

→ Convex shape haemorrhage

→ Injury to middle meningeal artery

MMA = Lucid

6 Sturge Weber Syndrome



→ Tram Track Calcification

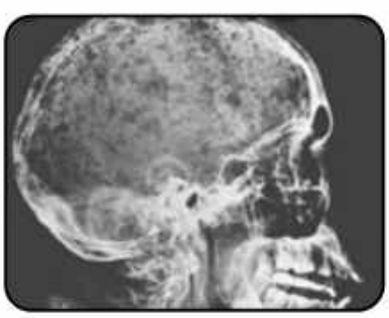
→ Port Wine Stain

7. Neurocysticercosis

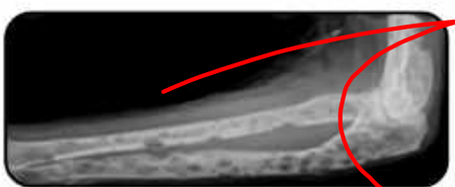
→ Parasitic infection by Pork Tape worm (Tenia Solium) = Tape worm..

→ On CT Scan = Cyst with a dot/ dot in a hole sign

8. Multiple Myeloma = Plasma cell cancer



→ On X-Ray = Rain Drop Lesion



→ Punch out lesion

9. Paget's Disease

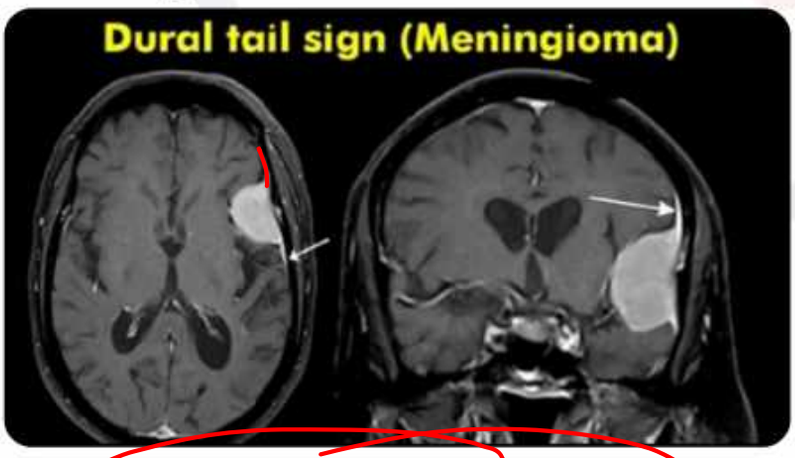
→ On X-Ray = Cotton wool skull

10. Diffuse Axonal Injury

→ Tiny Hypodense Foci on NCCT

→ Axon's are damaged

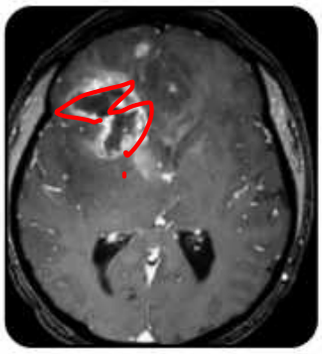
11. Meningioma



→ MC Benign Tumor

→ Psammoma Body

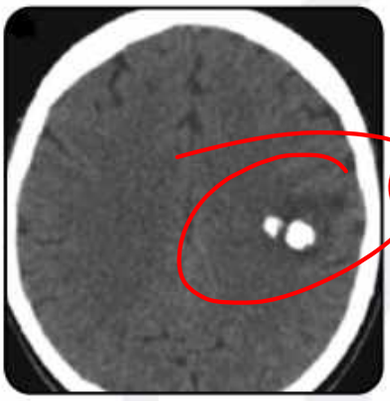
12. Glioblastoma Multiforme



→ Butterfly appearance seen on MRI

→ MC Malignant Tumor of Brain

13. Oligodendro Glioma



→ Ca+2 deposition seen on CT-scan

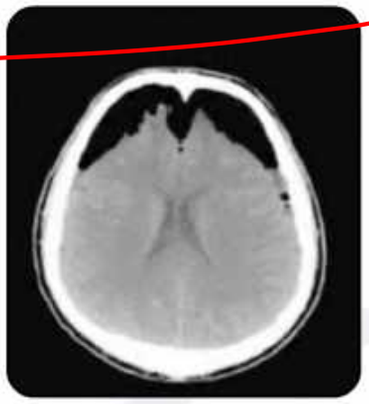
Fried egg cytoplasm →

14. Putamen Bleed

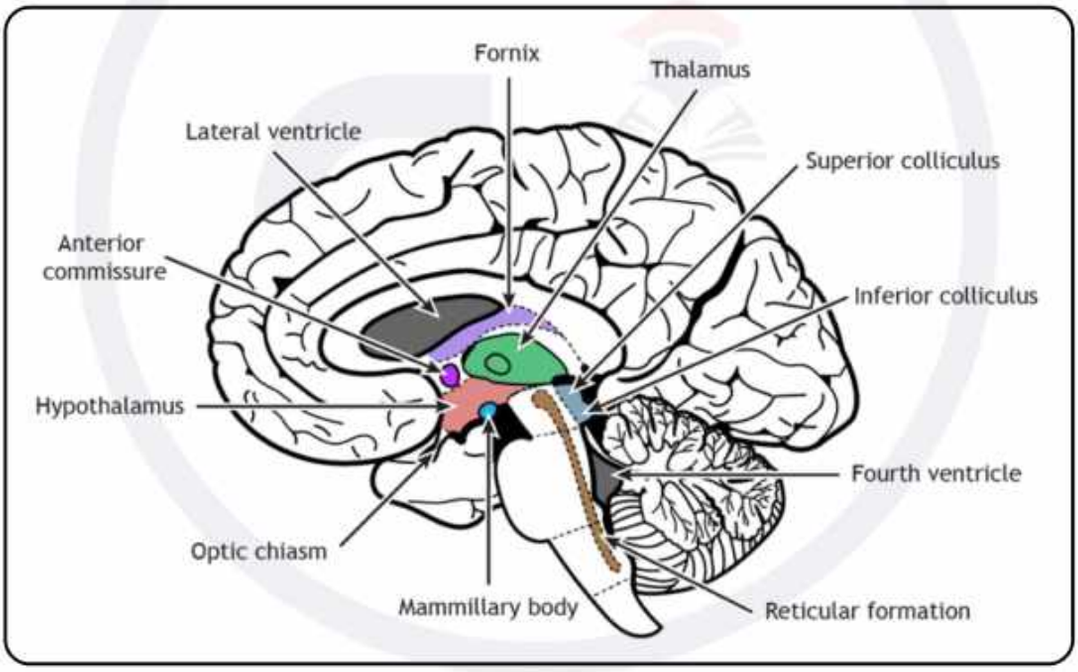
→ MCA

→ BG

15. Tension Pneumocephalus



→ Mount Fuji sign



Q1 → Sy in D/D in H?

Q- BRIDGING Vein = ? SDH

TIA + PWS ⇒ ?

SRFA → ?

Renal System

1. Renal Stone

→ Stag Horn Calculi = Triple Phosphate

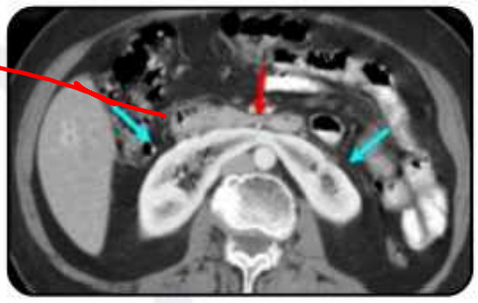
MC

2. Horse Shoe Kidney

→ Mc Renal Fusion Anomaly

→ Flower vase appearance seen

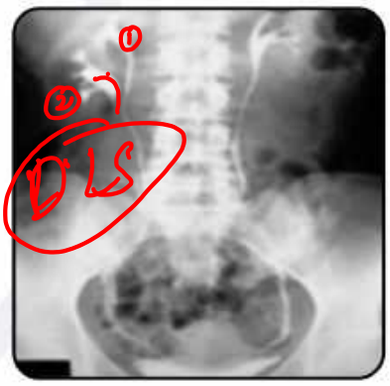
I.V.P



3. Duplex Pelvicalyceal System

→ Two ureter from kidney

→ Drooping Lilly sign is seen

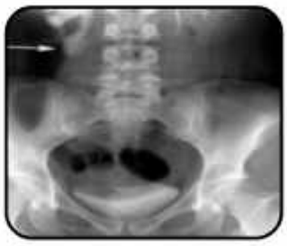


4. Ureterocoele

→ Cystic dilation of Vesico ureteric junction

→ Cobra head or adder head appearance is seen

5. Hydro Nephrosis



MCC = Pelvic Ureteric Junction Obstruction

6. Posterior Urethral Valve

→ MC - In male

→ On USG - Key hole appearance

→ Ix: MCU

7. Neurogenic Urinary Bladder

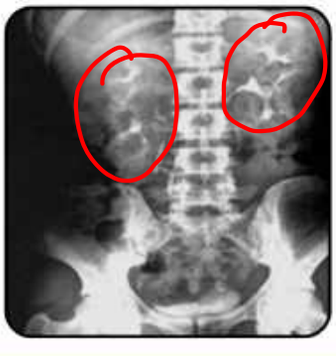
→ Seen due to increased sphincter tone

→ Christmas tree appearance

8. Polycystic Kidney Disease

→ Fluid filled cysts grows in kidney

→ Spider leg appearance is seen



CT Scan Abdomen

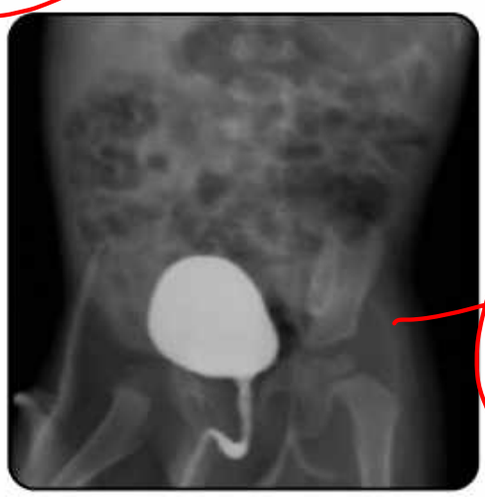
9. Renal TB

→ Putty Kidney appearance

C/P : Sterile Pyuria

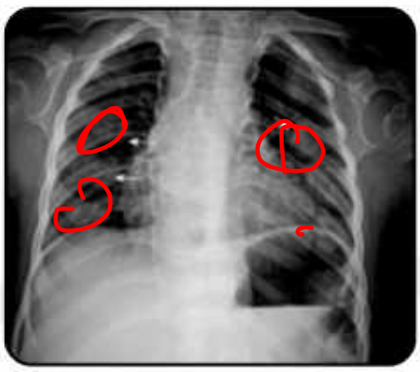
10. Medullary Nephrocalcinosis

11. MCU



Musculoskeletal System

1. Rickets = Disease due to Vit - D deficiency



Cuping

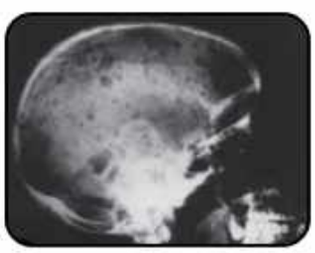
2. Scurvy =

→ Due to deficiency of Vitamin-C

→ White line of Frankle

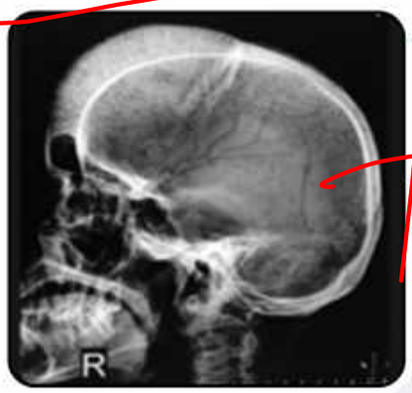
3. Hyperparathyroidism (↑ Bone Resorption)

→ Rugger Jersey Spine



Salt & Pepper Skull

4. Thalassemia =



→ Hair on end appearance/ Crew cut appearance

→

→ Chipmunk Face

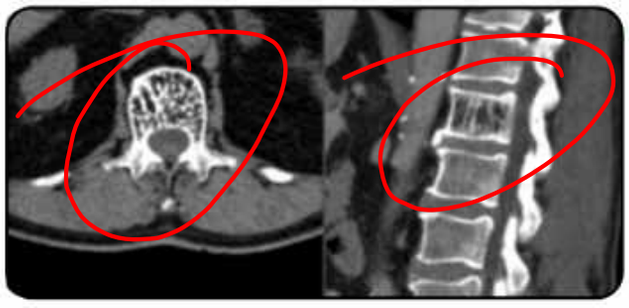
5. Sickle Cell Anaemia =

→ H -shaped vertebra

→ Hair on end appearance

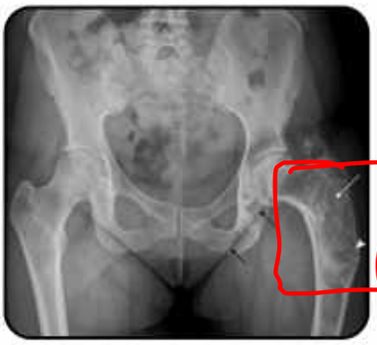
→ Spleen → H.S.B

6. Spinal Hemangioma



→ MC Benign Lesion in Vertebral Body

7. Fibrous Dysplasia



→ Ground Glass Matrix in Metaphyseal Region

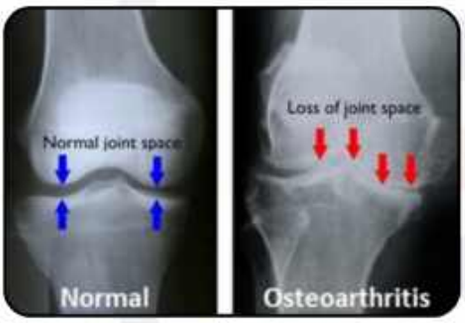
→ A/w Shepherd Crook Deformity

[Femur is Bend]

1-10

8. Arthritis

A) OsteoArthritis

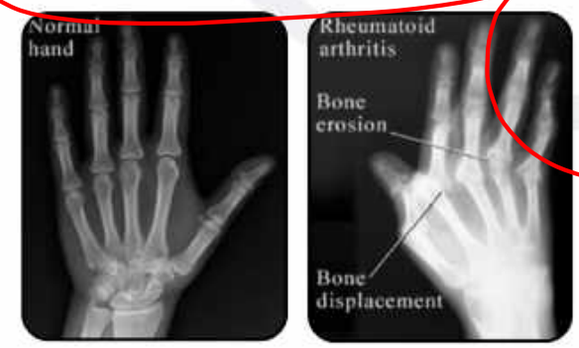


→ Degenerative Disease

→ Osteophytes seen (New Bone)

→ MC involved large joint at Lateral Side

B) Rheumatoid Arthritis



→ F>M

→ Small Joint > Large

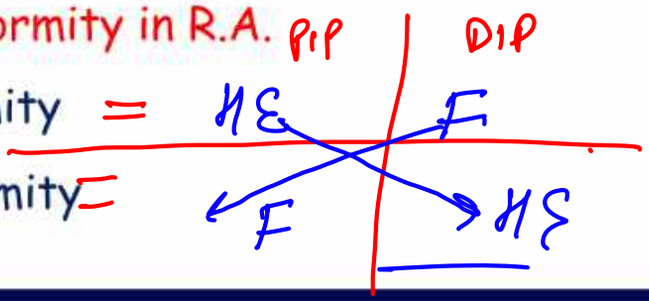
→ Morning stiffness +nt

→ MC Joint involved is MCP Joint [Periarticular Osteoporosis]

2 Most Important Deformity in R.A.

Swan Neck Deformity

Boutanniere Deformity



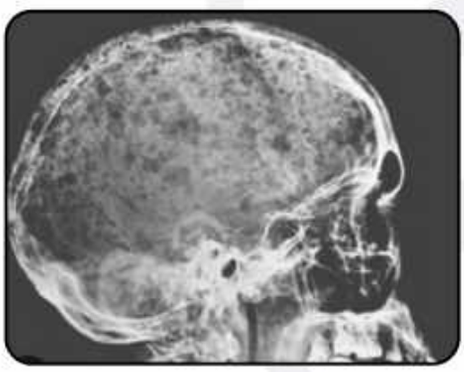
9. Charcot Joint/Neuropathic Joint

→ MC Seen in DM Patient

→ MC Organ involved is Foot

→ Painless condition

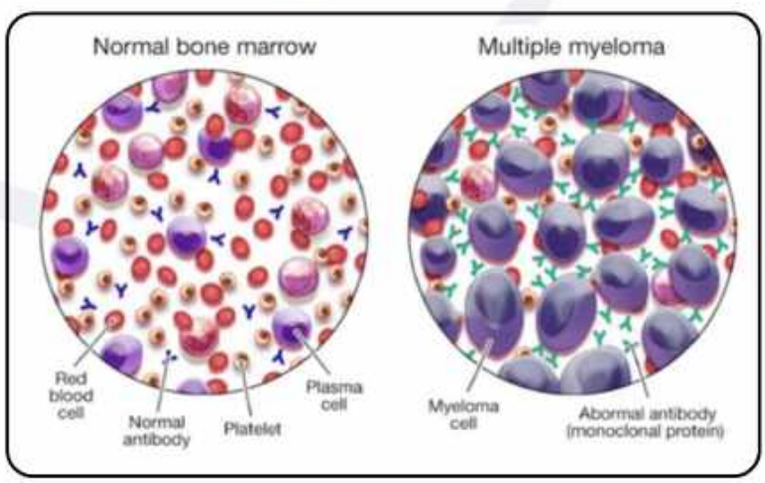
10. Multiple Myeloma/Kahler's Disease



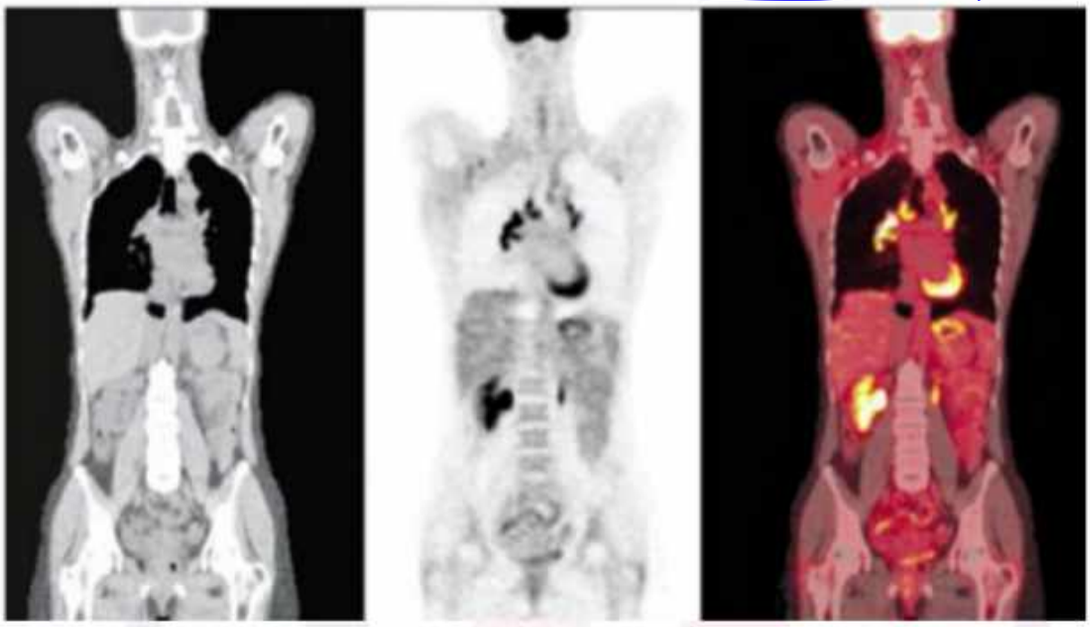
→ Multiple Rain Drop Lesion

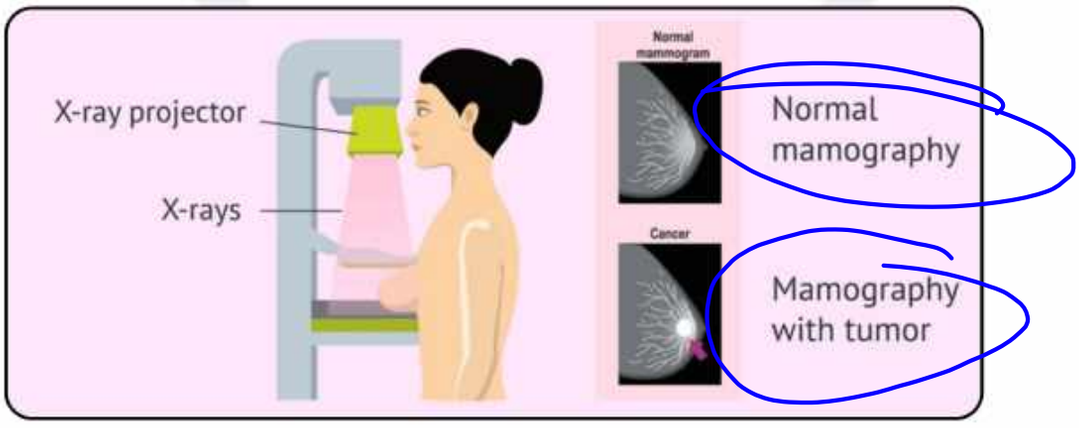
→ Plasma cell cancer

→ Bence Jones Protein seen in Urine

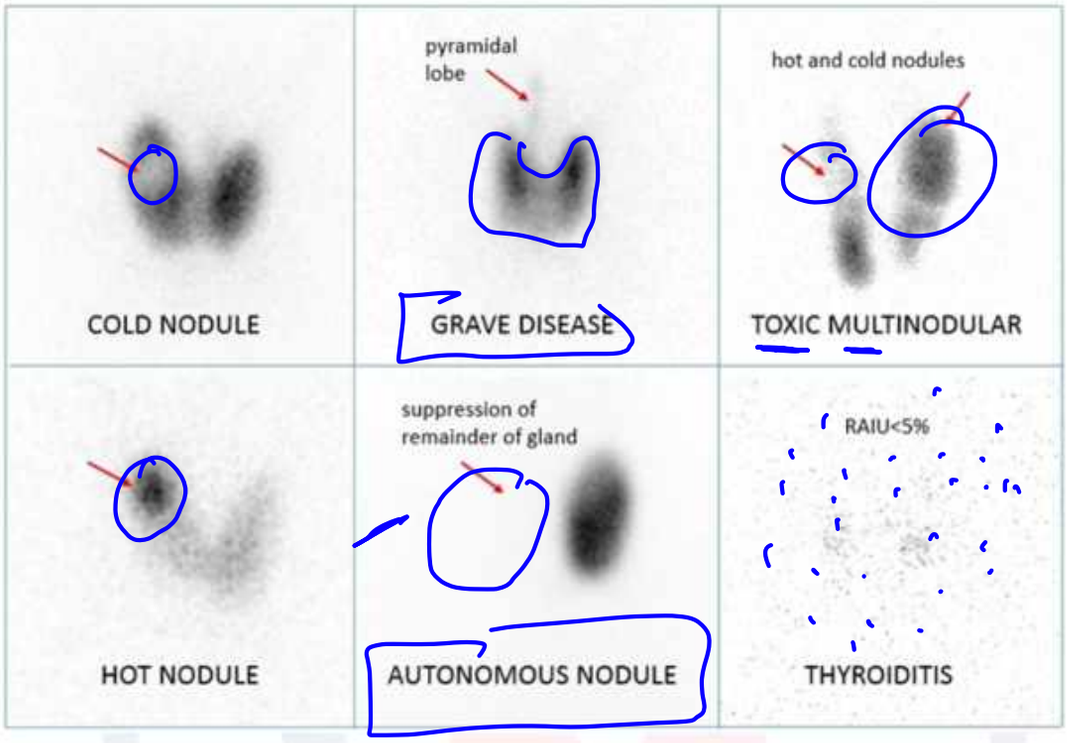


♦ Extra Edge

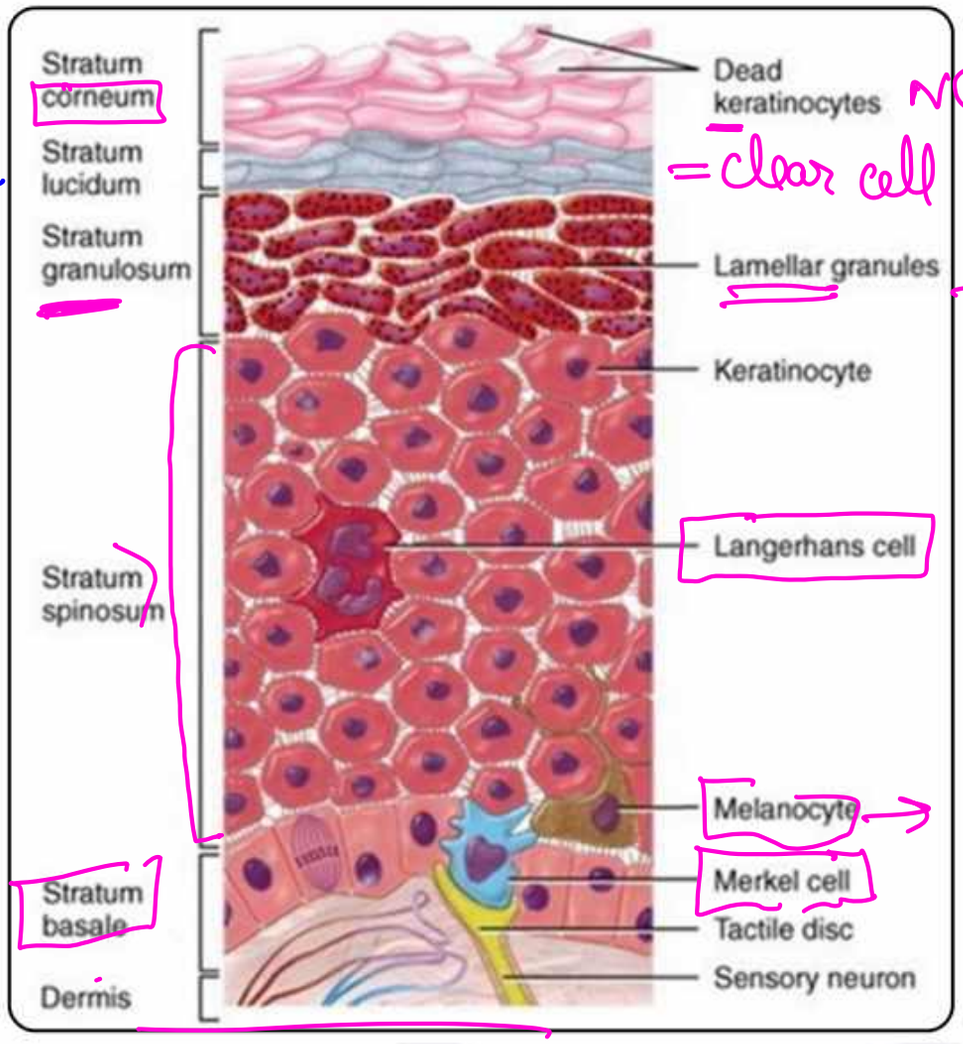

CT Scan
PET Scan
PET/CT Scan*


X-ray projector
X-rays
Normal mammogram
Cancer
Normal mamography
Mamography with tumor

♦ Technetium Scan



DERMATOLOGY



Cells	Layer	Derived from
1. Keratinocytes	All layer's	Ectoderm
2. Melanocyte	St. Basale	Neural crest cell
3. Merkel cells	St. Basale	Ectoderm
4. Langerhans cells	St. Spinosum	Mesoderm

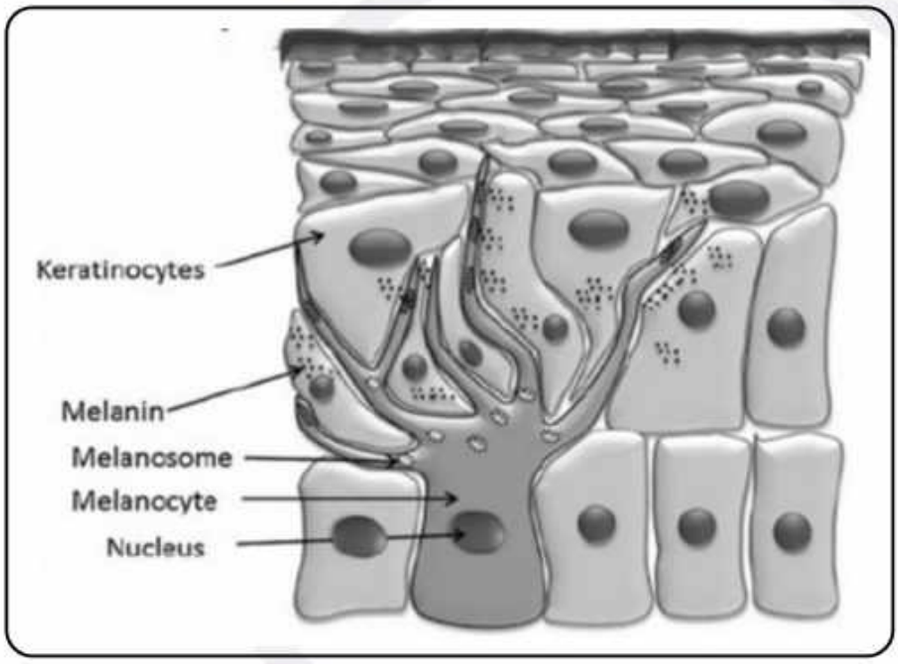
♦ Cells

1. Keratinocyte = 90%

- Hallmark – Presence of Keratin filament

2. Melanocyte

→ Synthesize - Melanosome which contain - Melanin



↓1 melan-
in unit
36 Keratinocytes Q

3. Merkel cell

- They are slow adapting type 1 touch receptor
- CK7-CK20 +ve

4. Langerhans cell

- Hallmark = Birbeck Granules
- APC → Phagocytosis
- CD207, CD1a +ve

♦ Layers[Q]

- St. Basale = 1st layer to be formed
- St. Spinosum = Thickest layer
- St. Granulosum = Contain filaggrin

↓ Deficiency

Winter Eczema

↓ Lamellar Bodies

↓

↓ Moisture of skin

↓

Lead to dry skin

Risk factors: Hot water bath

Rx.: Vit-D

Moisturizer +

Actinic Keratosis

- Cutaneous Horn
- Pre-malignant for Squamous cell carcinoma

- St. Lucidum = • Clear cell layer
  • +nt in palm & sole
- St. Corneum = • Lack nucleus
  • Contain dead Keratinocytes

Q1 Langhans → Dey ?

Q1 WE → ↓ ? LB

Q1 TML → P.

Blistering Disorder

- Blister = Fluid Filled Lesion's
  - ↳ < 0.5cm → Vesicle
  - ↳ > 0.5cm → Bullae

♦ Pemphigus

- Autoimmune Disorder
- IgG Ab Against Desmoglein → Blister's
- Erosion with crusting is seen

Pemphigus Foliaceus

= Flow.

- DSG-1
- St. Granulosum
- Superficial Bullae

Pemphigus Vulgaris

= Vul

- DSG-3
- St. Spinosum
- Superficial Bullae

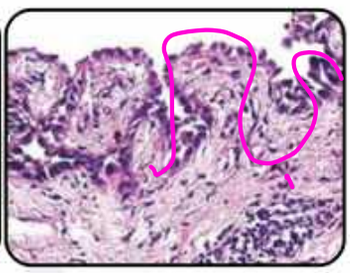
Deep

Clinical sign →

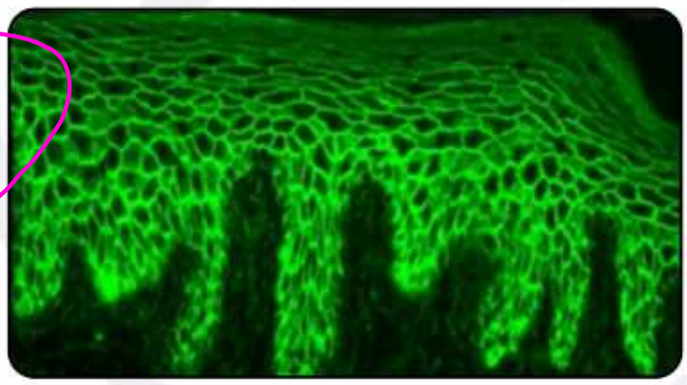
Tangential pressure

↓ Pressure → ↑ Size of Bulla

- Row of Tomb appearance



D/f =



→ Fish net like pattern
IgG + C3 deposit

Rx = Steroid
Cyclophosphamide
Rituximab

♦ **Bullous Pemphigoid**

- Ig-G Against BPAg2 Gene
- C/F - Tense Subepidermal Bullae →
- Blister +nt

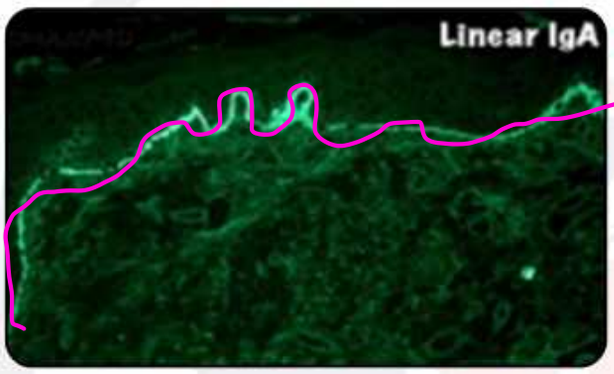
- Nikolsky Sign
- Row of Tombstone App

- nt

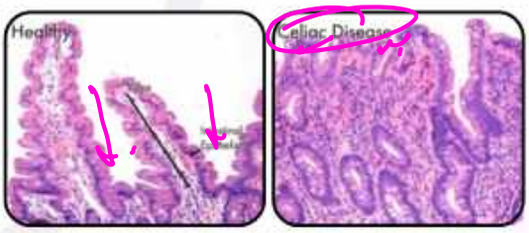
Rx.: Steroid

♦ Linear IgA disease

- IgA against BPAg2
- String of pearl or beads appearance
- D/f

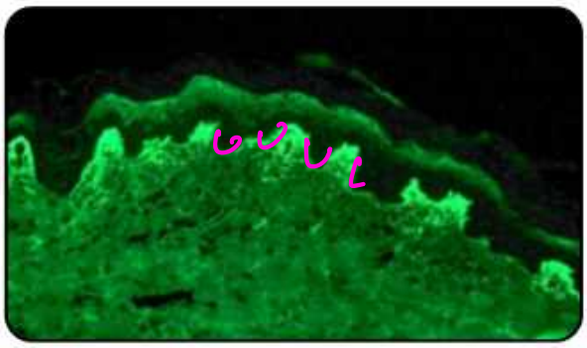


- Doc = Dapsone



♦ Dermatitis Herpatiformis

- Gluten Hypersensitivity
- DOC. = Dapsone
- TOC = Gluten Free Diet + Avoid → B - Barley, R - Rye, O - Oat, W- Wheat



♦ Darier Ds.

- AD
- Seborrheic areas
- Sand paper like app.

Rx. Oral Retinoids

Hailey-Hailey

- AD
- Skin folds
- Recurrent, Flaccid, Vesicular Lesions
- Brick wall app.

Rx. dipilumab (monoclonal Ab.)

♦ Erythema Multiforme

Cause = Viral infection's & drugs

Lesion = Target sign
Iiris lesion
Bulls eye

- Disease a/w Lyme's Disease

♦ Epidermal Lesion MCC = Drugs

- TEN - [Toxic Epidermal Necrosis]

BSA = >30%

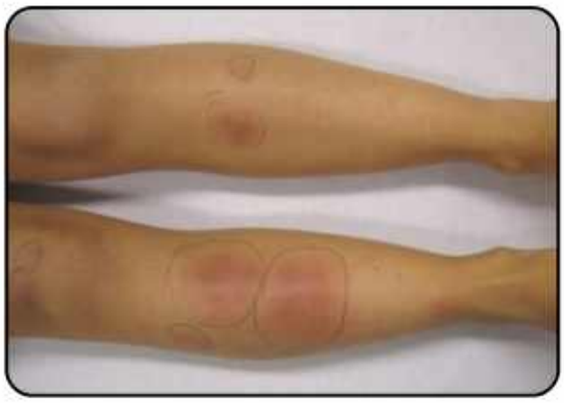
→ Pseudo-Nikolsky sign +nt

- Steven Johnson Syndrome

BSA = <10%

♦ **Erythema Nodosum**

- Inflammation of Subcutanous fat
- Tender lesion +nt



Cause: Sulfa drugs
Sarcoidosis

♦ **Erythema Marginatum**

- Reactive inflammatory condition associated with acute Rheumatic fever

Q ⇒ IgA aggn Desmoglein ?
IgA aggi. BPAG.
DOC → Dapsone
GMS = BPNS = ?
Blood STIES

Infection

1. Bacterial Infection

A. Erythrasma

- Caused by Corynebacterium Minutissmum
- Rash is +nt on axilla
- Dx - Wood's Lamp Examination - Coral red Colour

B. Non-bullous Impetigo

- Caused by Streptococcus pyogenes
- Mc - children & +nt on face

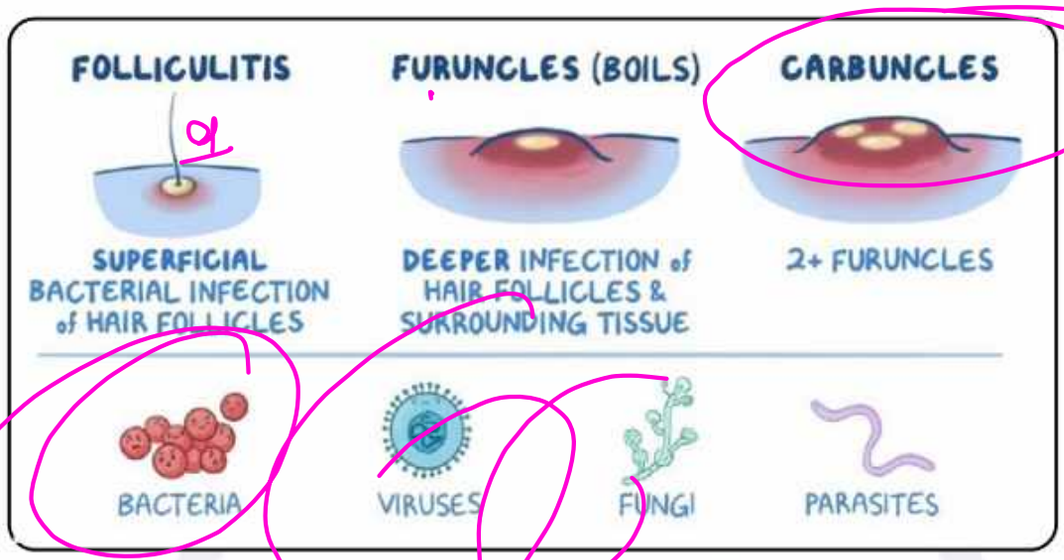
→ Golden yellow/ honey coloured crusting

C. Cellulitis =

- Caused by Streptococcus pyogens & Staph Aureus

→ Diffuse border's seen

D.



E. Erysipelas

Etiology: Group A Beta hemolytic streptococcus (Streptococcus pyogen's) Affect supereficial layers of skin

Rx: Penicillin P.O/I.M.

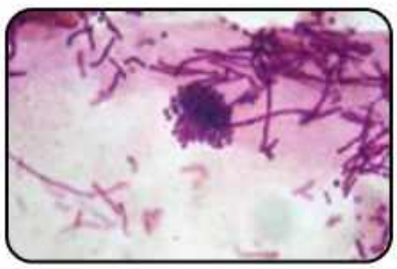
2. Fungal Infection

A. Pityriasis Versicolor

- Caused By = Malessezia Globosa > Furfur
- C/f = Rice Powder Like Scaling Is Seen
- Mc-Site = Upper Trunk

• Ix =

M. Furfur on SDA + Olive Oil = Fried Egg App.



10% KOH Mount

→ Spaghetti & Meatball

Woods Lamp

Yellow/ Apple Green Color

Rx = Topical → ~~Ketoconazole~~

Systemic → Selenium Sulphide

B. Candidiasis

↳ Vaginal → White curdy discharge

↳ Oral → Oral thrush

• Caused by → ~~Candida~~ Albicans

• DOC → Fluconazole

SDA = CPC ⇒ RBP

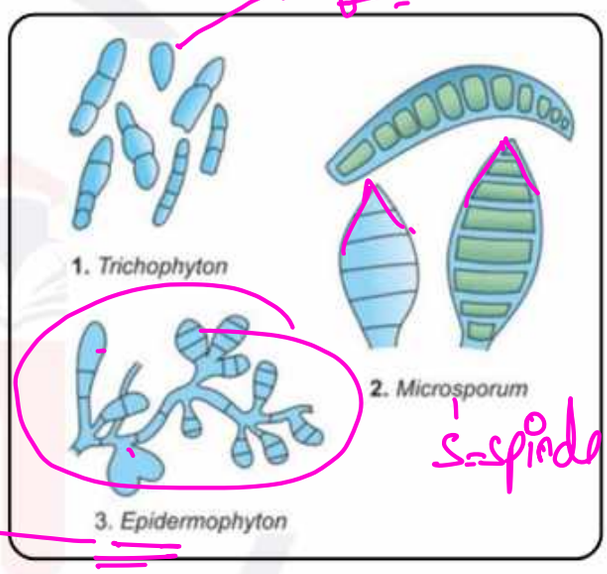
C. Piedra/ Trichomycosis Nodularis

→ Black Piedra → ~~Piedraia hortae~~

→ White Piedra → Trichosporon Beigeli

D. Dermatophytosis

	Skin	Hair	Nail
Trichophyton	+	+	+
Microsporon	+	+	
Epidermophyton	+		+



Epic = club = Boat = Pear

E. Tinea Infections

↳ Tinea Capitis = Head is involved

DOC: Griseofluvin

- Kerion

- Favus

↳ Tinea Manuum = Hands
↳ Tinea Pedis = Feet
↳ Tinea Unguium = Nail
↳ Tinea Cruris = Inguinal area

⇒ Terbinafine > Itraconazole

F. Sporotrichosis (Rose Gardner disease)

→ Caused By Sporothrix Shanki
→ Doc - Itraconazole
→ MC- Linear nodules along the draining lymphatics

Rx

G. Chromo Blastomycosis

→ Copper penny bodies
DOC → Itraconazole .

3. Mycobacterial Infections

A. Lupus Vulgaris
→ Plaque with central scaling
→ Diascopy = Apple jelly nodule

B. Scrofuloderma
→ Most commonly in children
→ Commonly +nt with sinus

Diagnosis

C. Leprosy

LL Type

Borderline Tuberculoid (BT)

- Erythemato-hypochromic plaque
- Dry surface
- Raised
- Well defined margins with satellite lesions
- Anesthetic

General Aspect of R-J Classification

Observation	TT	BT	BB	BL	LL
No. of lesions	Single usually	Single or few	Several	Many	Very many
Size of lesions	Variable	Variable	Variable	Variable	Small
Surface of lesions	Very dry	Dry	Slightly shiny	Shiny	Shiny
Sensation in lesion (Not face)	Absent	Mod - markedly diminished	Mod diminished	Slightly diminished	Not affected
Hair growth in lesions	Nil	Markedly diminished	Moderately diminished	Slightly diminished	Not affected
AFB in lesions	Nil	Nil or scanty	Moderate numbers	Many	Very many (globi)
Lepromin test	+ + + +	+ or + +	Negative	Negative	Negative

Virulence factor: PGL-1, LAM (Lipoarabinomannan)

MC Cranial nerve: Facial nerve

Overall MC nerve: Ulnar nerve

MC nerve for biopsy: Radial cutaneous nerve

Earliest sensation lost: Temperature (cold > warm)

Adult dose

Rx.: Rifampicin 600mg once a month
Clofazamine 300mg/month + 50mg daily
Dapsone 100mg daily

♦ Lepra Reaction

Inflammatory reaction occur in Leprosy

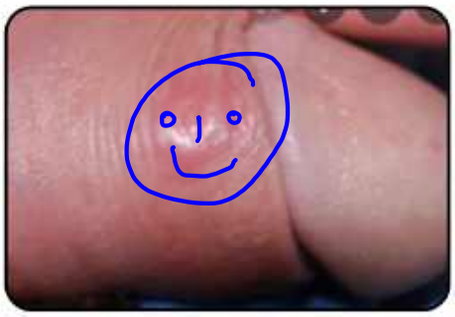
	Type-1	Type-2
• HSN	• Type-4	• Type-3
• Seen in	• BB/BT/BL	• LL
• C/F	• Pre-existing Tender raised Lesion	• New nodular lesion
• Systemic Involvement	• -nt	• +nt
• Treatment	• 1st NSAIDS • DOC: Steroid	• DOC: Steroid

STI (Sexually Transmitted Infections)

A. Syphilis =

PLC
RDP –
GF

- Caused by Treponema Pallidium
- C/F = Painless chancre +nt
- Rx = Benzathine Penicillin



Except Neurosyphilis → Aqueous Crystalline Penicillin

B. LGV = LGE Rho = Doctor

- Caused by - Chlamydia Trachomatis
- Rx → Doxycline

C. Chancroid = Painful Ulcer

- Caused by H. Ducreyi
- Rx Azithromycin

D. Donovanosis

- Caused By - Klebsiella Granulomatis
- C/F → Beefy red ulcers & bleed on touch
- Rx → Azithromycin

E. Congenital Syphilis

- Intestinal Keratitis
- Deafness
- Hutchinson's Teeth

F. Gonorrhoea

- Mcc - N. Gonorrhea
- DOC = Ceftriaxone

C/F → Purulent white discharge

Pain full U S I`
Pain less ,
q GS → BPV ?

Papulo Squamous Disorder

A. Psoriasis =

Mc - Psoriasis Vulgaris

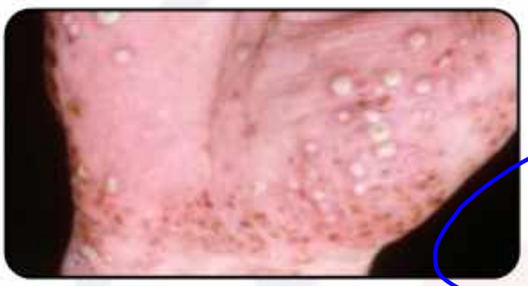
- C/F • Silver-White scaling
  - Auzpitz sign +ve
    (Grattage test)
- Type =
  → Plaque Psoriasis

Rx. Steroid

→ Guttate Psoriasis

Rain drop like Lesion
History of URTI
Etiology: Streptococcus
DOC: Macrolides

→ Pustular Psoriasis



History of Sudden Steroid withdrawal
Rx.: Acitretin

→Inverse Psoriasis

Rx.: Steroid

Treatment summary:

Rx → <10%: Topical Steroids, Retinoic Acid, Coal Tar
>10%: Phototherapy
↓ No response
Methotrexate

B. Lichen Planus

- Autoimmune condition

Oral Cavity: Lacy oral lesion

Associated with HCV

Nail: Nail Pterygium

Skin: Plane Purple Pruritic Lesoin

Koebner's phenomenon

- C/F – Koebner's phenomenon/ Isomorphic phenomenon

C. Pityriasis Rosea

Herald

christmas tree pattern

→Associated with HHV 6 &7

Rx- Self Limiting mostly

Low dose Steroids

D. Isomorphic/ Koebner phenomenon

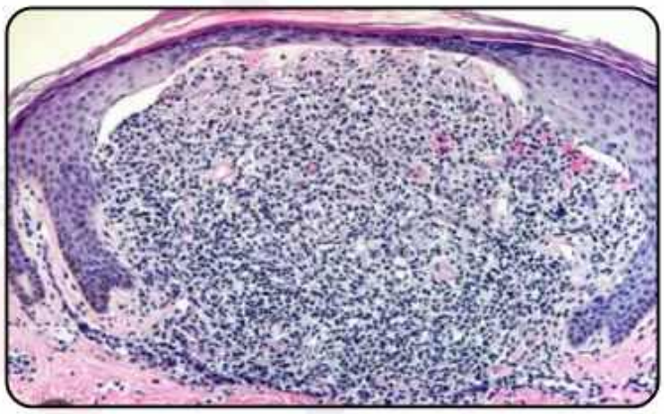
→+nt in Vitiligo
Psoriasis
Lichen Planus

Rx → Phototherapy

E. Lichen Nitidus

→Nodular Lesion

H/P - Claw Clutching a Ball Pattern



Pigmentary Disorder

A. Melasma = ↑Melanin

- Brown pigmentation seen after pregnancy

During Preg ⇒ CHOLASMA

B. Mongolian Spot

- Bluish spots over lower back

C. Nevus of OTA

D. Nevus of ITO

E. Becker's Nevus

- Hyperpigmentation + Hypertrichosis

F. Nevus of OTO

- Localized Depigmentation

Toxin		Toxin
Hydroquinone	=	Rubber
Parabutyl Phenol	=	Bindi
Parabutyl Catechol		Hair Dye

Queen chapr.

Hair

- **Chickenguniya** =? Aedes Mosquito

→Chick Sign due to Hyperpigmentation

Hx- Joint & Retro-orbital Pain

G. Acanthosis Nigricans

- Velvety skin
- MCC - Obesity

Mx-I = PCOS DM-II

H. Peutz-Jeghers Syndrome

STK11 Gene is involved

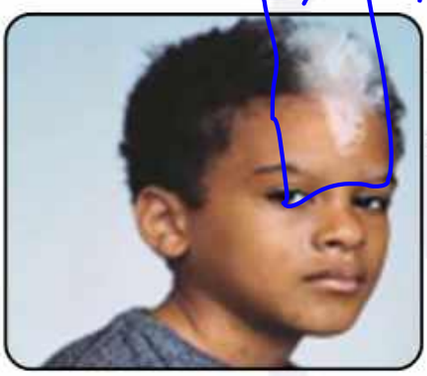
→Perioral lentigens seen

→Hamartomatous polyp in GIT

I. Fixed drug eruptions

- Hyper pigmented patch over lips/ genitals due to drug intake

J. Piebaldism



→Neural crest disfunction

→White forelock seen

K. Vitiligo

L. Freckles

M. Albinism

Skin Appendages

- Hair
- Nail
- Glands

♦ Hair cycle

I. Anagen → Growth = 3years
II. Catagen → Involution = 3 weeks
III. Telogen → Resting = 3 month

♦ Androgenic Alopecia

Mcc → DHT - Dihydrotestosterone

Male — Hamilton

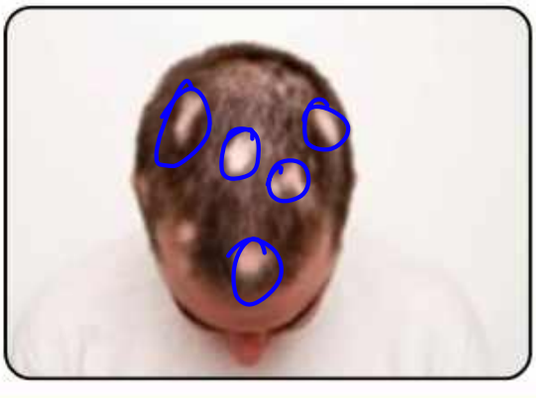
Female — Ludwig

Olsen

Rx = Topical → Minoxidil
Systemic → Finasteride

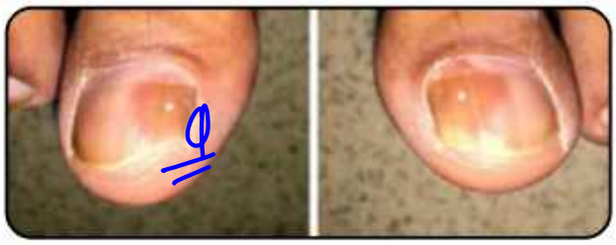
Sx = Hair Transplant

♦ Alopecia Areata



→Auto-immune condition
→Patchy hair loss
→Exclamation mark sign

Nail



A. Nail Psoriasis

MC-Characteristic finding

↓

Oil Drop Sign/ Salmon Patch

MC - Pitting

B. Terry nails

- Hepatic failure + CHF

C. Half & Half nails

- Chronic renal failure

D. Paronychia

- Infection of nail bed

E. Sub Ungal Hyperkera-tosis

- Thickening of nail plate

♦ Sweat glands

Eccrine

- Duct - Directly open
- Anhidrosis
- Hyperhidrosis
- Miliaria

Apocrine

- Duct-Open along hair follicle
- Bromhydrosis
- Chromhydrosis
- Fox fordyce disease

→Itchy papules are seen

♦ Acne Vulgaris

- Most Characteristic Feature
  - → Open = Black
  - → Close = White

Come Dones

Treatment →

Grade 1 → Topical Retinoid
Grade 2 → Topical antibiotic
Grade 3 → Oral antibiotic
Grade 4 → Isotretinoin

- **Rosacea**

→ Flushing seen after drinking hot liquids or **after working in hot areas**

Miscellaenous

♦ Capillary Hemangioma

→ +nt By Birth
→ Bright Red Plaque
→ Common Benign Neoplasm in Child

♦ Colloidon Baby

→ Congenital Ichthyosis
→ Where Baby Born With plastic like Membrane due to Abnormal Desquamation

♦ Harlequin Ichthyosis

→ Genetic Disorder [ABCA12]

♦ Staphylococcal Scalded Skin Syndrome

Etiology: Staph. Aureus

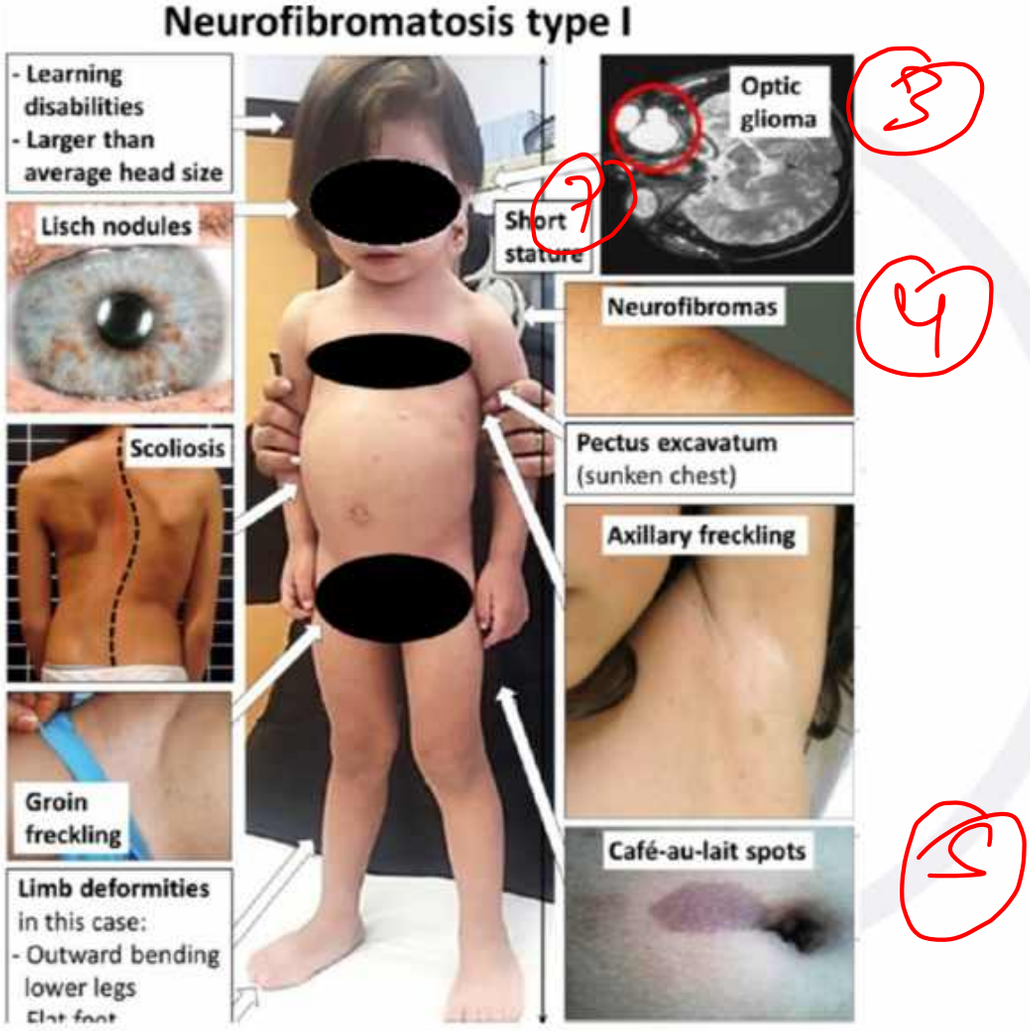
C/P: M.C. age 3-7 days of birth
Blisters conjuntiuitis

Rx: Antibiotics

♦ NF-1

→ Autosomal Dominant

→ Chromosome 17

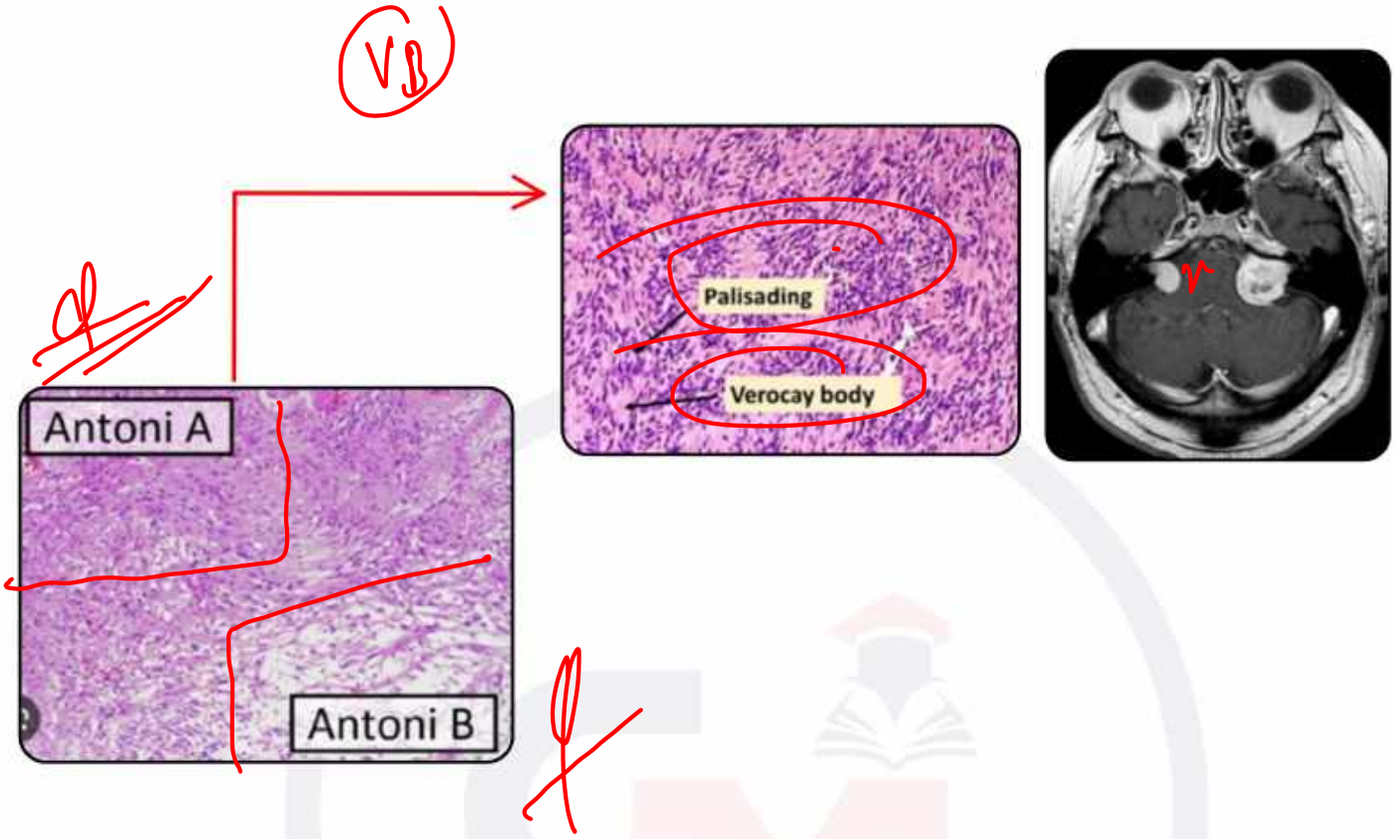


♦ NF-2

→ ~~Autosomal Dominant~~

→ Chromosome 22

→ Associated with Schwanoma



Q1 NF Type I ⟹ ③

NF1 → ?

Mildly → Hyperactivity ⟹ ADHD & Dyslexia

PEDIATRICS = 4.20.

♦ Jaundice

- Yellow pigmentation of skin • Sclera
- Yellow Urine Mcc → Bile Duct obstruction

Fetal Hb Brek

Iron

Heam
↓
Biliverdin
↓
Bilirubin → L.S → Unconjugated = Not Going.
↓ conj. by UDP Glucoronyl Transferase

Excreted In Urine bcz. Water soluble ← conj. Bilirubin
↓ [Comes Out Through B.D]
Secreted In Bile → Intestine
→ Stercobilin → Yellow Brown Faeces

Acholuric/ Prehepatic
- Skin & Sclera Yellow
- Urine normal ~~-N~~

Choluric/ Post Hepatic Jaundice
- ~~Urine Yellow~~
- ~~Skin yellow~~ +ve.

Physiological	Pathological
• Onset after 24 hrs	• Before 24 hrs
• Last < 7-14 day's	• >7-14 days
• Daily rise - <5mg %	• > 5 mg%
• Total Bilirubin < 15mg%	• > 15 mg %
Tx → Reassurance	• Tx Phototherapy • Phenobarbitone

Exam

Kramer Rule → Visual assessment of bilirubin level

→ Head → 5mg %
→ Chest → 10mg %
→ Abdomen & Thigh → 12-15mg %
→ Palm & Sole → 15-20mg %

♦ Neurology

- Seizure Disorder → ↑ Electrical activity in brain
- Convulsion → Electrical activity + Motor activity
- Epilepsy → > 2 Seizure episodes
  - ↳ Mc Epilepsy → Subtle seizure
- Mcc of seizure → HIE [Hypoxic Ischemic Encephalopathy]

Type →

1. Absence Seizure
   → Doc - Ethosuximide in children < 2yrs
   Valproate > 2yrs
2. GTCS
   → Doc - Valproate
3. Febrile Seizure
   → I.V Lorazepam
   → Rectal Diazepam
   → Prophylaxis - Clobazam

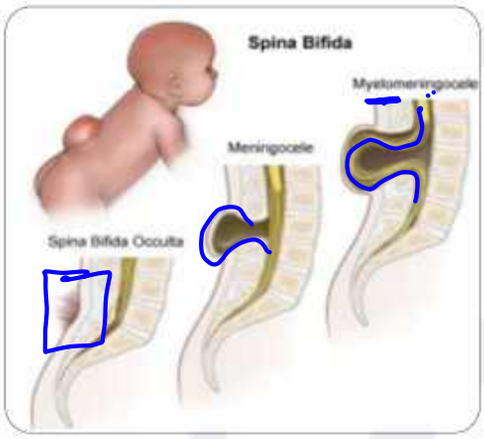
♦ Neural Tube Defect

Dx → Most Specific → Cholinesterase
Most Sensitive → AFP
Prevention By → Folic Acid = Dose ? = 500mcg.

NTD → - 4000mcg.

9

Spina Bifida Occulta	Meningocele	Myelomeningocele
→ Hair & skin covering defect	→ Meninges comes out	→ Meninges & nerve fibres comes out



Brain

Encephalocele

♦ **Dehydration**

→ Mcc in children → Diarrhoea

→ Mcc Of diarrhoea → Rotavirus → E. Coli

Mild to moderate	Severe
→ Child is thirsty	→ Child is drowsy
Rx - ORS → Na^{2+} - 75ml/kg	Rx - I.V - R.L = 15kg/h
Glucose - 75	1st 10 kg 100ml/ kg
K^+ - 20	2nd 10 kg 50ml
Citrate - 10	30% 70%
Cl^- - 65	Age <1yr - 1hr 5hr - 6hrs
	Age >1yr - 1/2hr 3/2hrs - 3hrs

♦ CVS

•Fetal Circulation

→ Blood supply → 2 Umbilical artery & 1 vein (It)

→ Ductus Venosus → Left umbilical vein → Inf. Vena Cava

→ Ductus Arteriosis → Pul. Artery → Arota

→ Foramen Ovale → R. Atrium → L. Atrium

Congenital Heart Disease

Cynotic : Blue	Asynotic : Pink
R → L shunt	L → R shunt
T → TOF	D - ASD - Mid Systolis & Diastolic
T → TGA	D - VSD - Pan Systolic Murmur
T → TAPVR	D - PDA - Contnous Murmur
T → Tricuspid Atresia	
T → Truncus Arteriosus	
E → Ebstein Anomaly	
E → Eisen Menger phenomenon	

♦ Dx → CHD

[Congenital Heart Disease]

NADAS CRITERIA	
MAJOR CRITERIA	MINOR CRITERIA
SYSTOLIC MURMUR Grade 3OR MORE	SYS. MURMUR LESS THAN Grade 3.
DIASTOLIC MURMUR	ABN.SECOND SOUND
CYANOSIS	ABN. ECG
CHF	ABN. CxR
	ABN. BP

♦ Neonatology

→ Weight at birth → 2.8-3 kg
→ Low birth weight → < 2.5kg
→ Head circumference → 35cm
→ If >2cm ↑ → Hydrocephalus
→ At birth → HC > CC
→ After 1 Year → CC > HC

- New-born Do not shiver
  Do not sweat
  Generate heat by lipolysis of brown fat
- Hypothermia < 36.5 °C
- Hyperthermia > 37.5 °C
- New Born Loose Heat
  1. Conduction - Contact with cold surface
  2. Convection - Source of cool air
  3. Evaporation - Loss through skin [Mc at birth]
  4. Radiation - Near but not in touch with cold surface
     ↳Overall Mc
- Temp. Of NICU - 25°- 28°

♦ KMC

→ Indication to start - <37wks or <2.5kg
→ When to stop - >37 wks or >2.5kg

♦ Sequency of Puberty →

Girls = 1st - Thelarche
Last - Menarche

Boys = 1st - Testicular growth
Last - Beard/ Spermarche

♦ Best method for growth assessment in children → ↑ Weight & Height

↳ Weight

Normal = 2.8-3 kgs

X 2 = 5months

X 3 = 1 year

X 4 = 2 years

↳ Length

At birth = 50cm

3 months = 60cm

9 months = 70cm

1 year = 75cm

2 years = 90cm

4 years = 100cm

♦ Mid arm circumference

→ Bangle Test

→ Shakir Tape → Arnold's classification

R Y G

- Normal - >12.5
- Mild - 11.5-12.5
- Severe - <11.5

Malnutrition

Duration

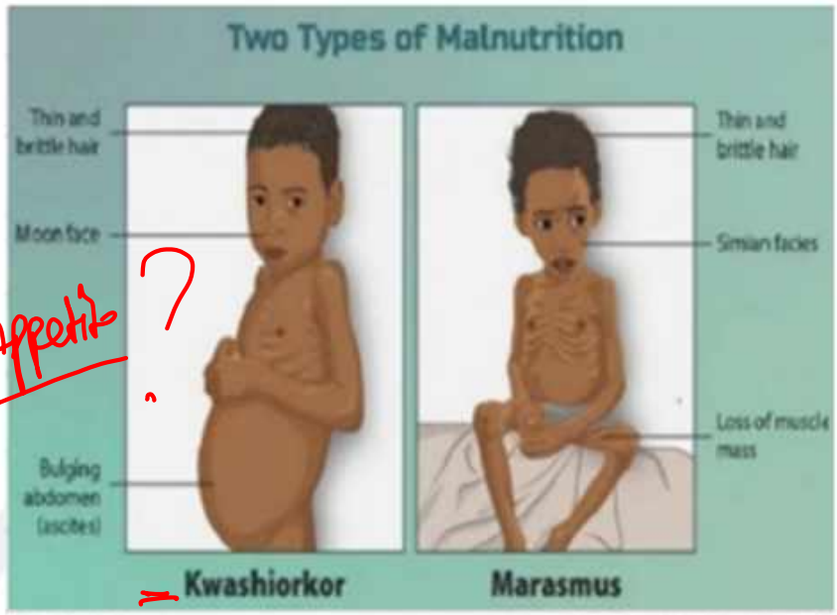
Acute
- Weight is Reduced
- [Wasting]
- Chart
- Wt. for HT

Chronic
- Height is Affected
- [Stunting]
- Chart
- HT for age

Based on severity

Severe Malnutrition
- B/L Pedal Edema symmetrical
- Wt. for HT < -3 SD
- Height for age <-3 SD

	Kwashiorkor	marasmus
Predominant deficiency	Protein	Calories
Onset	1-4 years	< 1 year
Edema	+++	Absent
muscle wasting	+ (hidden)	+ visible severe
Appetite	Poor	Good
General appearance	Lethargic, apathy	Alert
Additional features	Hair changes (flag sign). Skin changes. Presence of fatty liver	


Two Types of Malnutrition
Thin and brittle hair
Moon face
Bulging abdomen (ascites)
Thin and brittle hair
Simian facies
Loss of muscle mass
Kwashiorkor
Marasmus

♦ Nutrition

- Breastfeeding → Start ASAP
- Exclusive breastfeed → Upto 6 months

↳ Prolactin Reflux

- Production of milk → Galactopoesis

↳ Oxytocin Reflux

- Ejection of milk → Galactokinases

- Cow milk vs Human milk

	Cow milk	Human milk
Carbs	↓	↑
Protein	↑	↓
Fat	Same	Same

- Storage temp of breast milk

→ In Room Temp → 6-8 hrs

→ In Refrigerator → 24 hrs

→ In Freezer (-20°C) → 3 months

Short Stature

- **Physiological**
  - **Familial**
    - Short
    - Ontime
    - → Final HT
    - → Puberty
  - **Constitutional**
    - Normal
    - Delayed
- **Pathological**
  - Stunting (Malnutrition)
  - Turner Syndrome
  - Down Syndrome ET
  - **Disproportionate**
    - US/ LS ratio Ab for age
    - **US > LS** [Short Limbs]
      - → Achondroplasia
      - → Rickets
      - → Cong. Hypothyroidism
    - **LS > US** [Short Trunk]
      - → Spondylo Epiphyseal Dysplasia
      - → TB = POTT SPINE
      - → Vertebral Abnormalities
  - **Proportionate**
    - US/LS ratio normal

- Overall Mc type of short stature
  - = Constitutional delay
- Mcc Pathological Short Stature
  - = Malnutrition
- US/ LS - Ratio
  - → AT BIRT = 1.7:1
  - → 2 years = 1.5:1
  - → 7-10 years = 1:1

- Fetus pass urine by - 1st 24 hrs
- Fetus pass meconium - 24-48 hrs
- Primitive Neonatal Reflexes

PRIMITIVE REFLEXES

REFLEXES	DESCRIPTION	AGE REFLEX APPEARS	AGE REFLEX DISAPPEARS	
Rooting	When infant's cheek is stroked, the infant turns to that side, searching with mouth.	Birth	3 months	
Sucking	Reflexive sucking when nipple or finger is placed in infant's mouth.	Birth	2-5 months	
Moros	With sudden extension of the head, arms abduct and move upward and the hands form a "C"	Birth	4 months	

4- Parachute Reflex

Appears - 6-9 months after birth & persist

- **Apgar Score**

Apgar score

	Score 2	Score 1	Score 0
Appearance	Pink	Extremities blue	Pale or blue
Pulse	> 100 bpm	< 100 bpm	No pulse
Grimace	Cries and pulls away	Grimaces or weak cry	No response to stimulation
Activity	Active movement	Arms, legs flexed	No movement
Respiration	Strong cry	Slow, irregular	No breathing

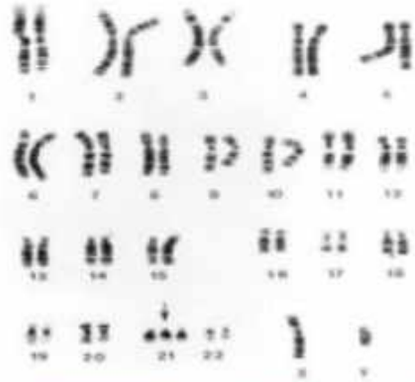
- <3 = Severely depressed
- 4-7 = Moderate
- >7 = Normal

= PINK = 120 - 2
= Weak Pulse 1
= Good Grimace 1
= Strong Cry 2

= 8

Term	Preterm
Dry, Pink skin	Smooth, Shiney skin
Instant ear recoil	Slow ear recoil
Lanugo -nt	+nt
Breast bud 5-10mm	<5mm

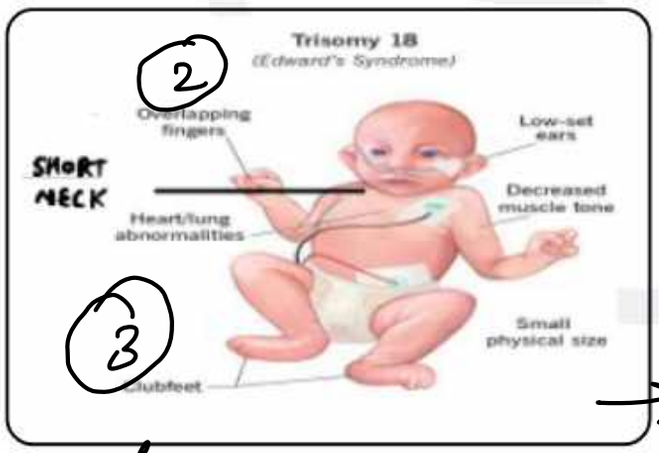
Genetic Disorder



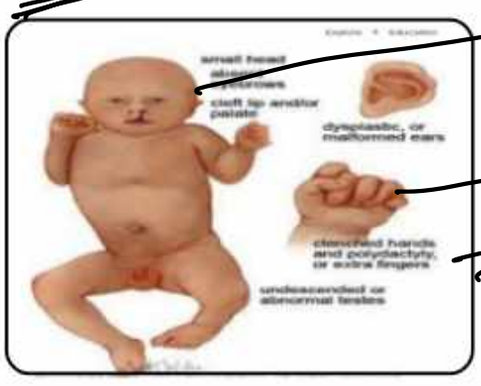
1. Down syndrome = Trisomy 21

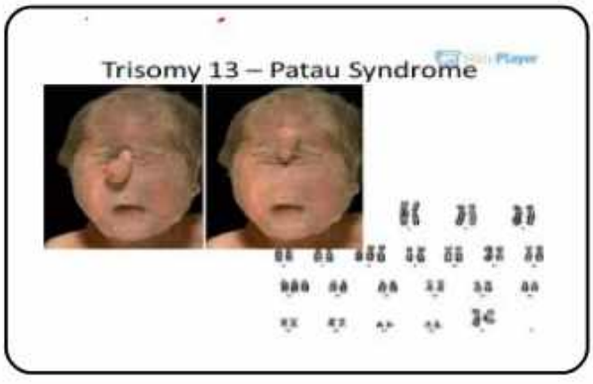
→ Mc genetic cause of low IQ

2. Edward syndrome = Trisomy 18

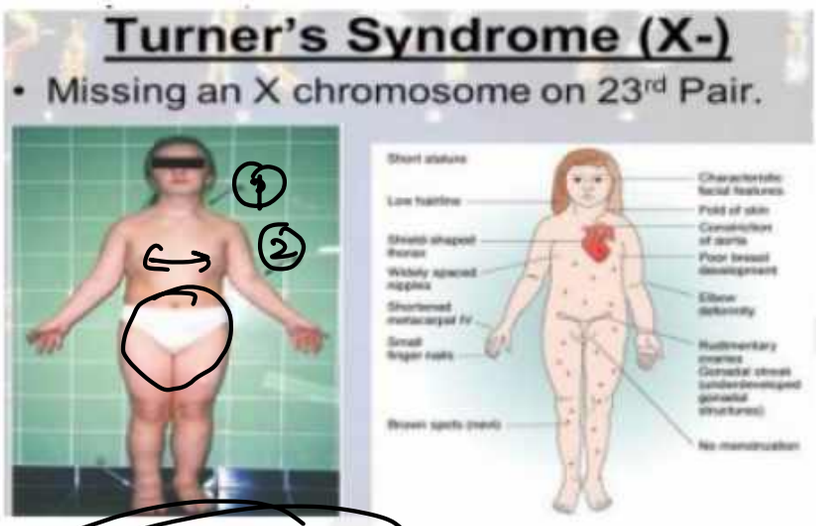


3. Patau Syndrome = Trisomy 13





4. Turner Syndrome = 45×0



→ Seen in females
→ Normal IQ

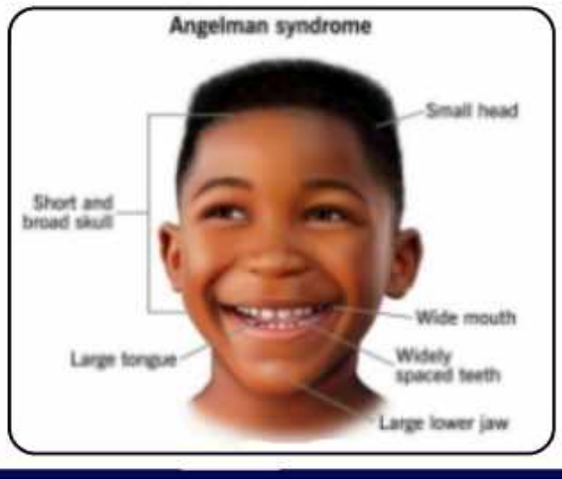
5. Noonan Syndrome/ Pseudo Turner Syndrome

→ Seen in both M & F
→ Normal karyotype 46XX or 46XY
- Undestended testis & triangular face
→ Low IQ
→ Rest all features are same as turner

6 Klinefelter Syndrome = 47XXY

→ Long leg
→ -nt axillary & pubic hair
→ Small Penis & Testis
→ Puberty starts at time & progress slow

7. Angelman Syndrome



↓IC
→ Genetic condition - 15q chromosome
→ Very low IQ
→ Smiling face
→ ↑ seizures

♦ Mile Stone

1. Language Milestone

- → 6 Month - Mono-syllabus
- → 9 Month - Bi-syllabus
- → 1 Year - 2-3 Words
- → 2 year - Sentence
- → 4 Year - Story

2. Social Milestone

- → Social smile - 1.5 months
- → Recognise mother - 3 months
- → Mirror smile - 6 months
- → Bye Bye - 9 months
- → Peek A Boo - 10-12 months
- → Stranger anxiety - 7 months
- → Separation anxiety - 15 months
- → Colour, gender, names - 3 years
- → Undress - 2 years
- → Dress - 3 years
- → Left or right hand - 4 Years
- → Bowl control → 4yrs
- → Bladder cont. → 5yrs

3. Fine Motor Milestone

- → 4m - Bi-Dextrous approach
- → 5m - Transfer
- → 6m - Undexterous approach
- → 7m - Ulnar Grasp
- → 8m - Radial Grasp
- → 9m - Immature Pincer Grasp

→ 12m - Mature Pincer Grasp
→ 15m - Cup without spill
→ 18m - Spoon without spill/ scribbling
→ 2yrs - Line
→ 3yrs - Circle
→ 4yrs - Square
→ 5yrs - Triangle
→ 6yrs - Hexagon

4. Gross Motor Milestone

	Support	Without support
Sit	6m	8m
Stand	9m	12m / 1 year
Walk	12m	15m
Run		18m

2yrs - Jump + Climb stairs
4yrs - Skipping & climb stairs on alt foot

→ BYE - BYE =

→ Sepration Anxiety =.

→ Recognize mother =

→ Mature Pincer =

→ Stand without support =.

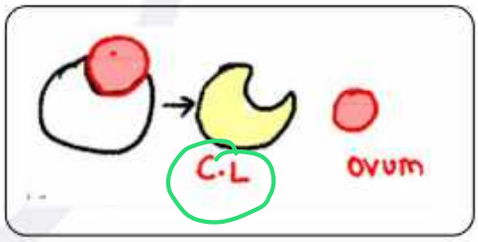
→ Peak a Boo Game =

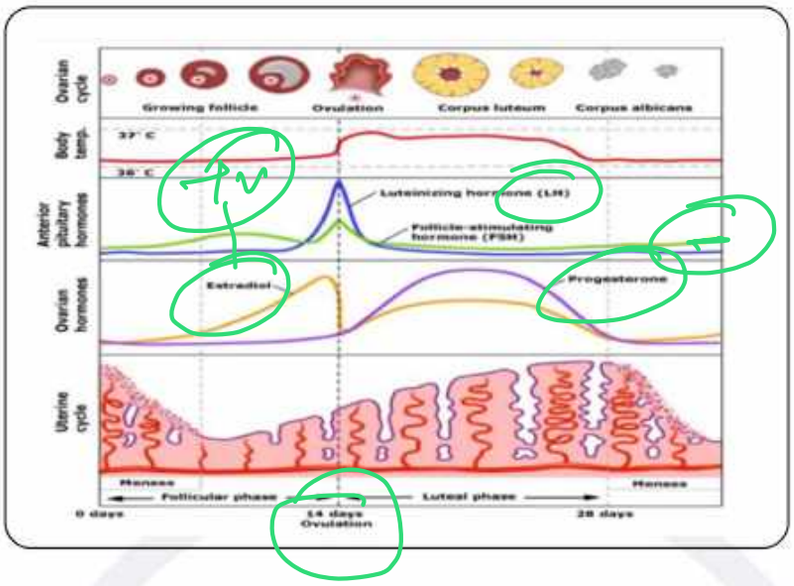
S.20 PJAMWAL7

Gynaecology

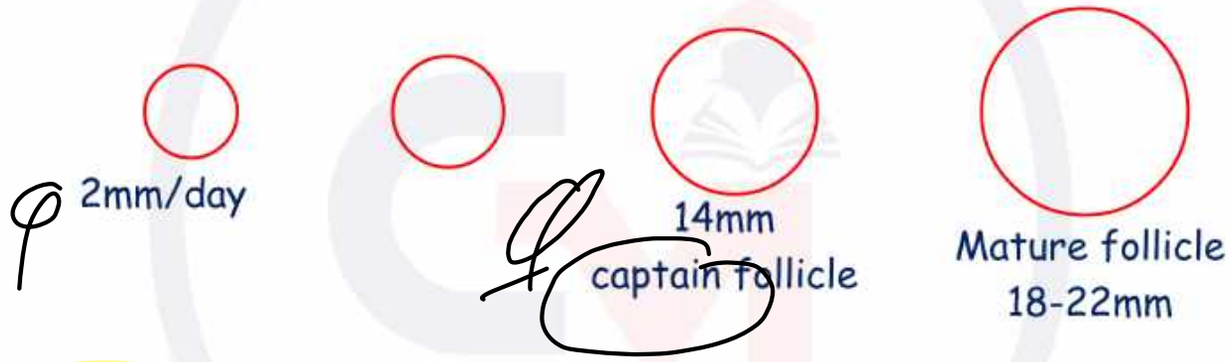
♦ Menstrual Cycle

Hypothalamus Secrete GNRH
↓
GNRH acts on anterior pituitary
↓
Anterior Pituitary Secrete FSH
↓
FSH acts on ovarian granulosa cells
↓
Estrogen production
↳Helps in proliferation
↓
If Estrogen level > 150-200pg
It will give +ve feedback to LH
↓
LH & FSF is secreted by - Pituitary Gland
↓
LH ↑ [14^{th} day] - Ovulation
↓
Corpus Luteum Secrete Progesterone
↳Progesterone make uterus secretory
↓
If progesterone level >3mg/ml gives -ve feedback to LH
↓
LH ↓ → Corpus Luteum degenerate
↓
↓ Progesterone
↓
Endometria shedding occurs





- **Follicular growth**



- LH peak → 10-12 hrs before ovulation
- Min LH required for ovulation - 75mg/ml
- Min Estrogen req. for LH surge - 200pg/ml

♦ Mittelschmerz Pain

Mid-Cycle pain
↓
Advice USG
↓ If normal
Rx - Reassurance

♦ Mittelschmerz Blut

Mid Cycle spotting
↓
Advice USG
↓ If normal
Rx - Reassurance

- How to calculate day of ovulation [DOO]

  Example 35 days cycle

  DOO - 35-14 = 21st day

- Normal menstrual cycle length
  - → 21-35 days
  - → Average - 28 days

  Blood Loss - 5-80ml

- Normal duration of bleeding → 2-7 days

- Abnormalities Of Menstruation

1. Oligomenorrhea
   - → Delayed menses (>35 days)
   - → Mcc - PCOD
2. Polymenorrhea
   - → Early Menses (<21 days)
   - → Mcc - PID
3. Menorrhagia
   - → Excessive menses - (>80ml or > 7 days)
   - → Mcc - Uterine Fibroid
4. Metrorrhagia
   - → Irregular Bleeding

   Mcc - Dub - Dysfunctional Uterine Bleeding
   [Without Organic Pelvic Pathology]

   AUB - Abnormal Uterine Bleeding
   [With Organic Pelvic Pathology]

P → Polyp
A → Adenomyosis
L → Leiomyoma (Fibroid)
M → Malignancy
C → Coagulation Disorder
O → Ovulatory Disorder
E → Endomaterial Disorder
I → Iatrogenic
N → Not yet classified

5. Hypomenorrhea
Less Menses - <5ml or < 2 days
Mcc- Asherman Syndrome [Adhesion in Endometrium]

6. 1° Dysmenorrhea
→ Pain during menses
→ Doc - Mefenamic acid

2° Dysmenorrhea
→ Mcc - Endometriosis

7. Amenorrhea
→ Absence of menses

A → Physiological Amenorrhea
→ In Pregnancy → Due to HCG
→ In Lactation → Due to Prolactin
B → Pathological Amenorrhea

I. 1° Pathological Amenorrhea	II. 2° Pathological Amenorrhea
Never Menstruated	Absence of menses for 3 cycle in previously normal menstruating women

I. 1° Pathological Amenorrhea

A. At Endometrial Level

- → Cervix
- → Uterus
- → FT
- → Upper part of vagina

1. Mullerian Agenesis
   - → Absence of Mullerian duct
   - → Ovary normal
   - → She can have own baby by IVF
   - → 2° Sexual character normal

MRKH - Mayer Rokitansky Küster Hauser Syndrome

Mullerian agenesis + Renal abnormalities + Spinal abnormalities

2. TFS - Testicular Femnisation Syndrome/
   AIS - Androgen Insensitivity Syndrome
   - → External genetalia
   - → Karyotype - 46XY
   - → 2° sexual character like female
   - → Mistaken as female at birth
   - → Axillary hair & pubic hair absent

Q. Best clinical method to differentiate TFS & mullerian agnesis - Absent Axillary Hair

Q. Best method - Karyotyping

B. At ovarian level

1. Turner Syndrome

   45 x 0

Short stature - ↓ Growth hormone

Streaky Ovary - ↓ Estrogen

Rx - Estrogen + Progesterone

2. Pure Gonadal Dysgenesis

46 XX

Height normal

Some part of X chromosome is not formed properly

Rx - Estrogen + Progesterone

3. Sweyer Syndrome

→ 46XY

→ External genetilia is of male

→ Bcz of mullerian duct - Internal organs are of female

Rx - Vaginoplasty at marriage

Orchidectomy asap

C. At Pituitary level

1. Craniopharyngioma

→ ↑ Prolactin level

Δ MRI

Rx - Cabergoline

↓

Sx

D. At Hypothalamus level

1. Kallman Syndrome

→ Mutation in KAL-1 gene

→ Infertility + Anosmia

R → Pulsatile GnRH

1° → EM → ① MA. 2° ✓
② TFS ✗

→ Ovary - ① Turner
② PCD
3 Sw ← m / XX (E)

→ Pitu - ① Gro

II. 2° Pathological Amenorrhea

A. At Endometrial level

1. Asherman syndrome

→ Fibrous band ~~nt which inhibit~~ proliferation

Rx → Hysteroscopic removal

B. At Ovarian Level

1. Polycystic Ovarian Syndrome

C. At Pituitary Level

1. Pituitary Tumor

2. Sheehan syndrome

→ Postpartum necrosis of anterior pituitary D/T Pph.

Rx - Substitution of all hormones

D. At Hypothalamus

→ Due to stress & excessive wt loss

Asherman — 1°

Sheehan — ? Sweeby

PGP

♦ Puberty

↳ Age of development of 2° sexual character in female 8-12years [10.5 yrs]

I. Precocious Puberty

↳ 2° sexual character develop < 8 Years

Mcc - Idiopathic

Rx - Continuous GnRH

II. Precocious Menarche

↳ Menses begin < 10 years

Mcc - Idiopathic

Rx - Continuous GnRH

III. Delayed Puberty

↳ 2° sexual character do not develop > 13 yrs.

Mcc - Idiopathic constitutional delay

Rx - Pulsatile GnRH

♦ PCOS / Stein Leventhal Syndrome

→ Path-physio

↳ Insulin resistance → Hyperglycaemia → ↓ Sex hormone binding globulin → ↑ Free Androgens → ↑ FSH → Small Follicles 2-9 mn X >= 12 in no.

Symptom

→ Tendency towards obesity

→ Acne, Hirsutism, Androgenic Alopecia

→ Oligomenorrhea > Amenorrhea - Infertility

Long Term Complication

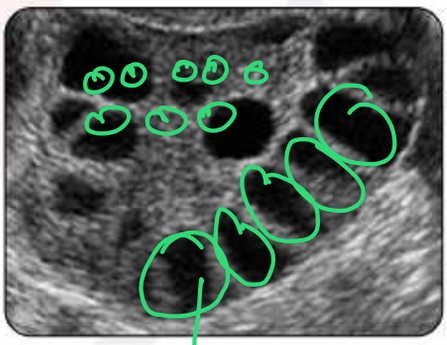
→ Metabolic X syndrome [Obesity, DM, HTN] → Skin = Acantho

Dx Criteria - Rotterdam Criteria

→ Menstrual Dysfunction

→ ↑ Androgen - 70-150ng/ml

→ USG - Size - 2-9mm
No - ≥ 12
App - Necklace sign



OHSS

→ E2: E1 - 1:2

→ FSH: LH - 1:2

Mx - Life style modification

↓

Doc - DM - Metformin

Doc - For androgenic feature - OCP

Doc - Oligomenorrhea - Progesterone withdrawal bleeding

Doc - Infertility - Letrozole

♦ Vaginitis

1. Trichomonal Vaginitis

→ Caused by - Trichomonas Vaginalis

→ Symptom - Greenish white discharge + Itching + Dysuria [↑ Frequency of Micturation]

→ Sign - Tiny ulcers on vagina

↳ Strawberry vagina
↳ Angry looking vagina
↳ Colpitts macularis

→ Ixoc - Saline Microscopy

→ Mx - Tab - Metronidazole

Male partner is treated

2. Candidal Vaginitis

→ Caused by - Candida Albicans

→ Symptom - Curdy white discharge from vagina + Intense itching + Dysuria

→ Ixoc - Saline microscopy

→ Mx - Fluconazole - 150mg - Single Dose

→ 1st Trimester → No Rx.

→ 2 & 3rd Trimester → Clotrimazole - Cream

Male partner is treated

3. Chlamydial Vaginitis

By - Chlamydia Trachomatis

!! Most Deadly Std

Ixoc - NAAT - Nucleic Acid Amplification Test

Gold Standard - Mc-Coy/ Hella cell culture

Mx - Azithromycin
Doxycycline
+ Contact Tracing

4. Bacterial Vaginosis

→ By - Gardnerella Vaginosis

→ Doderlein Bacili ↓ → PH > 4.5

Acid → Baj

→ Diagnosis - by Amsel Criteria

- pH > 4.5
- Thin Greyish Discharge
- Clue Cell > 20%
- +ve Whiff Test/ Amine Test
  - ↳10% KOH + Discharge → Fishy Odour

Gold Standard - Nugant Score - > 7

Rx - Metronidazole

Q1 ALV = ? = DOC = METRONIDA, Ca?

Cu

♦ PID - Pelvic Inflammatory Disease

→ Inflammation of uterus, fallopian tube, ovaries

→ Mc organism in acute case - Neisseria Gonorrhoea

→ Mc organism in IUCD user - Actinomycis

→ Mcc - Multiple intercourse τ multiple partner

• Acute PID

→ C/F - Lower abdominal pain, fever

- Pyosalphinx

→ Diagnosis

1. Minimum criteria

Cervical, Uterine & Adenaxal tenderness

On USG

2. Additional criteria

→ Temp > 101°F

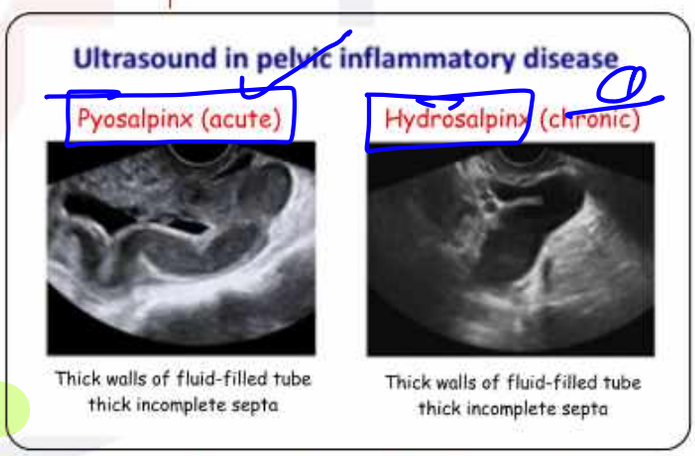
→ ↑ WBC, ↑ CRP

3. Definitive criteria

→ Endometrial Biopsy

→ USG - Tubo Ovarian Mass

→ Laparoscopy - Sunset sign [Trickling of pus from fimbriae]



→ Management

OPD	IPD
• Inj. Ceftriaxone [Gonorrhoea]	• Inj. Clindamycin
+	+
• Tab. Doxycycline [Chlamydia]	
+	
• Tab. Metronidazole	• Inj. Gentamicin

- Chronic PID

C/F - Dull pain on movement
- Hydrosalpinx

Ix - Laproscopy - violin string app.
↓
Dx - Fitz Hug Curtis syndrome

Complication
↳ If tube is completely blocked - Infertility
↳ If tube is partially blocked - Ectopic Pregnancy

♦ Endometriosis

Presence of endometrial tissue outside uterus

Mc-site - Ovaries

R/F - ~~• K-Ras Gene~~
• High Estrogen
• Nulli Parity

Protective Factor
→ Smoking - ↓ Aromatization
→ OCP - ↑ Endometrial Thinning

Symptom - Dysmenorrhea [Menstrual pain]
Dyspareunia [Intercourse pain]
Infertility

P/V Exam - Adenexal Mass

USG - Ovarian Cyst - Ground Glass appearance

IXOC - Laparoscopy - Chocolate cyst on ovary

Gold Standard - Histopathological examination

Management

→ Ovarian Cyst

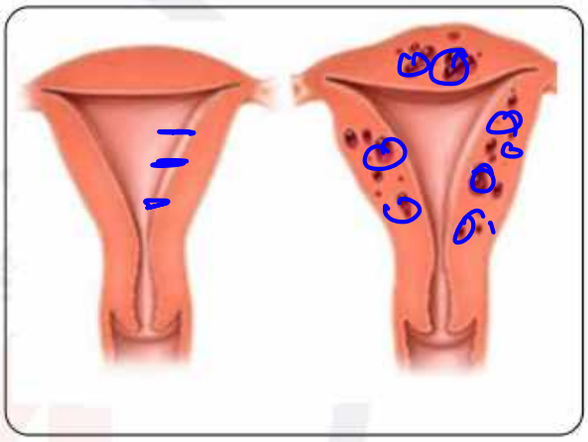
↳ < 3cm - Cyst aspiration

↳ ≥ 3cm - Cystectomy

→ Endometrial deposit - Cauterisation & Fulguration

→ Adhesion - Adhesiolysis

→ Doc - Cont. GNRH



♦ Adenomyosis

→ Endometrium in myometrium

R/F - Multi Parae

Age - ≥ 40years

Mc Symptom - Menorrhagia

USG - Salt Pepper appearance

IXOC - MRI

Gold Standard - Hysteroscopic Curettage

Mx - Total Abdominal Hysterectomy

EMJ Thickness.

♦ Fibroids

Origin - Myometrium

R/F - ↑ Estrogen, nulliparity
Genetic translocation Chr. 12 & 14
High socio-economic class

Protective Factor - Physical exercise
Smoking

Uterine Fibroid -

1. Intramural - Mc
2. Sub Mucous
3. Subserous

Extra Uterine Fibroid

1. Cervical Fibroid
   - Ant. Wall → ↑ Frequency of urine
   - Post. Wall → Urine retention

Mc Symptom - Menorrhagea

IxOC - USG

Best Ix - MRI

Complication
→ Hyaline degeneration Mc
→ Calcareous degeneration
→ Red degeneration

Rx -

If Asymptomatic [Size < 10cm]
↳ USG follow up to check size

If Symptomatic
↳ Multiparae → TAH
↳ Nulliparae → Myomectomy

Medical Mx - Cont. GnRH
+ Tranexamic acid

♦ Congenital malformation of Uterus

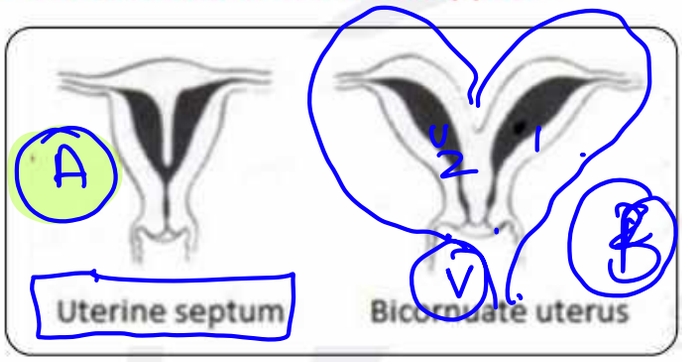
→ Mullerian duct start developing - 7-8 wks of IUL

→ Mullerian duct start fusing - 12 wks

→ Mullerian Duct → Uterus, Cervix, Fallopian Tube, Upper 1/3 of vagina

- Abnormalities

→ If inner wall don't disappear



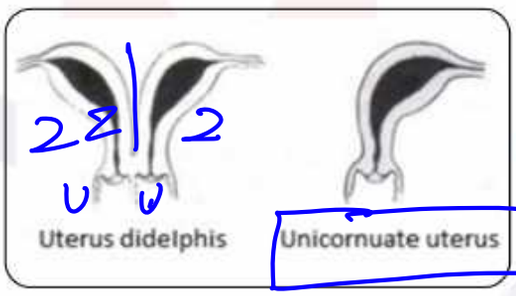
→ Heart shape uterus

→ Md fuse partially

→ 1 Vagina + > 2 Cervix + 2 Uterus

→ Md do-not fuse

→ 2 Vagina + 2 Uterus



→ Only on MD develop

Ix - Mc - USG

Gold standard - Laparoscopy + Hysteroscopy > MRI

Rx -

1. Septate Uterus - Hysteroscopic resection of septum
2. Uterine Didelphys/ Bicornuate - Metroplasty

♦ Infertility

→ Failure to conceive even after 1 year of unprotected & regular intercourse

Female Factor

I. Ovarian Factor

1. H-P-O - Axis failure

- Kallmann Syndrome Rx - Pulsatile GnRH

2. H-P-O axis dysfunction

- PCOS - Rx Letrozole

- Hyperprolactinemia - Rx Bromocryptine

II Tubal Factor

→ Mcc - PID > Endometriosis

Best test to check tubal patency → Laparoscopic Chromopertubation [Best day 10]

Rx Of Tubal Block - IVF

III Cervical Factor

Mc - Anti Sperm Ab.

Dx - Sim's Or huhner Test → See Jerky pattern

Rx - Intra Uterine Insemination

Male Factor's

Normal seminogram by WHO

→ Amount - Min 1.5ml

→ Sperm Count - >= 15million/ml

→ Total Sperm Count Per Ejaculation - 39 million/Ej.

→ Morphologically Normal Sperm - ≥ 4 %

→ Viable Sperm - ≥ 58%

→ pH - 7.2

→ Total Motility - ≥ 40%

Q1 Semen < 1.5ml
- Semen
m NS < 4%

Abnormalities

→ Semen < 1.5ml - Hypospermia

→ Semen > 5.5ml - Hyperspermia

→ Semen -nt - Aspermia

→ Sperm -nt - Azoospermia Q

→ Sperm Count < 15million/ml - Oligospermia Q

→ Morphologically normal Sperm < 4% - Teratospermia Q

→ Motility < 32% - Asthenospermia Q

→ Viable Sperm < 58% - Necrozoospermia

Male Factor

→ Pre testicular

→ Testicular

→ Post testicular

Pre Testicular	Testicular	Post Testicular
→ Hypothalamus	→ Accident	→ Obstructive
→ Pituitary	→ H/O Radiotherapy	
FSH ↓	N	Normal
LH ↓	N	Normal
Testosterone ↓	↓	Normal
Mx - Pul. GnRH	Mx - Accordingly	Mx -

- TESA [Testicular Sperm Aspiration]
  - → OPD
- (Best) MESA [Microsurgical Epididymal Sperm Aspiration]
  - → IPD
    - If fails
      - [IUI] Intra Uterine Insemination

- Indication of IUI
  - → Impotence
  - → Vaginismus
  - → Mild Oligospermia
  - → Hypospadias
  - → Retrograde Ejaculation

9

- ART [Assisted Reproductive Technique]
  - → IVF - In vitro fertilization
  - → ICSI - Intra cytoplasmic sperm injection
  - → ZIFT - Zygote intra fallopian transfer
  - → GIFT - Gamete intra fallopian transfer

Q - IUI is not ART

♦ Menopause

→ Cessation of menses for at least - 1 year

→ Menopause age in India - 47 years

→ Menopause age in World - 51 years

- Premature Menopause - Before 40 yrs
  - ↳ Mcc - genetic
- Delayed Menopause - > 55 years
  - ↳ Mcc - Uterine Fibroid

Pathophysio → Cessation of HPO axis

↓

↓ Estrogen, Progesterone

↓ Endometrial Thickness

↑ FSH, LH ↑

Symptom's & Treatment

→ Hot Flushes - Doc Tibolone

→ Osteoporosis - Doc Bisphosphonates

→ Dry Vagina - Estrogen cream

→ ↓ Libido - Androgens

- Premature Menopause - < 40 yrs

Dx - Anti mullerian hormone [< 1ng/ml - Ab]

S.FSH on day 3 [>40-45 IU/L] Ab

Obstetrics

♦ Placenta

→ Weight - 500gm

› Maternal component → Formed by Decidua Basalis

› Fetal Component → Formed by chorion frondosum

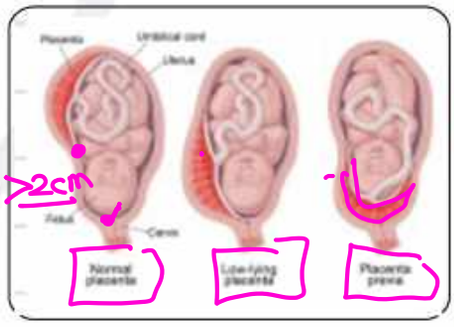
→ Site - Upper Posterior/ Fundo Posterior

↳ Hormones Secreted by Placenta

- hCG By synchiotrophoblast cells in placenta
- Estrogen
- PAPPA [Pregnancy Associated Plasma Protein - A]
- HPL [Human Placental Lactogen]

• Distance b/w lower edge of placenta & internal Os - ≥ 2cm

↳ Abnormalities

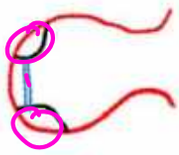


1. Low Lying Placenta

→ Lower edge is within 2cm from internal Os

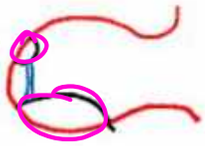
2. Accessory Lobe

A. Placenta Bilobata



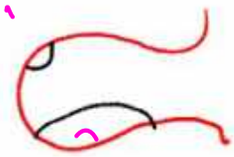
→ Two equal size lobe's connected blood supply

B. Succenturiate Placenta



→ One small & one big lobe connected by blood vessel

C. Placenta Spuria

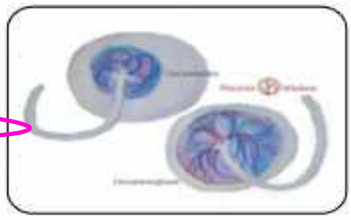


→ 1 Small & 1 big lobe not connected by blood vessel

3. Extra choral Placenta

Maternal surface larger than fetal surface

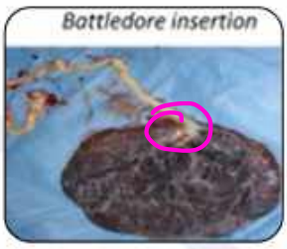
A. Circum Marginate Placenta



B. Circumvallate Placenta

4. Battledore Placenta

→ Umblical cord is attached at placental margin



5. Vilamentous insertion of cord

→ Umbilical cord is inserted on fetal membrane & then reaching placenta

- How to differentiate b/w maternal & fetal blood
  - ↳ Singer Alkali Denaturation test/ APT test
  - ↳ Reagent - 10% KOH
  - ↳ Interpretation - Fetal Hb is resistant

♦ Amniotic Fluid

Source

- < 7 wks - Ultra filtrate of maternal plasma
- 7-20 wks - Diffuse from fetal skin
- > 20 wks - Fetal urine

Turn Over Time - 500 c.c/hr

Normal

- Single Deepest Pocket - 2-8cm
- Amniotic Fluid Index - 5-25cm
- Amount - 200-2000ml

Oligohydroamnion	PolyHydroamnion
• SDP < 2cm	> 8cm
• AFI < 5cm	> 25 cm
• Volume < 200ml	> 2000ml
• Mcc Of Mild - Idiopathic	- Idiopathic
• Mcc Of Severe - MDH [Maternal Diastolic HTN.]	- Open Neural Tube Defect
Rx Amnio infusion of normal saline	Amniocentesis

- Max amount of amniotic fluid is +nt in
  - 32 Weeks - 34 Weeks - 1 litre
  - 36 Week - 900ml
  - 42 Week - 200-400ml

	O	I
SDP	< 2	> 8
AFI	< 5	> 25
AF	< 200	> 2000

Colour of amniotic fluid

- → AT Term → Straw colour
- → RH Incompatibility → Golden yellow
- → Listeria Infection → Green
- → Abruptio Placenta → Dark colour
- → Fetal Demise → Blackish brown
- → Post Term Pregnancy → Greenish yellow [Saffron colour]

Pregnancy

→ Peroid Of Amenorrhoea - 40wk/ 280 days

→ POA is calculated from 1st day of LMP

Growth Period

→ Ovular Period → Upto - 2 week

→ Embryonal Period → 3-8 week

→ Fetal Period → 9 week onwards

Trimester

→ 1st Trimester → upto 14 weeks

→ 2nd Trimester → 14-28 week

→ 3rd Trimester → 29wk - 42wk

GTPAL Scoring System

G → No. of times of pregnancy

T → No. of term deliveries

P → No. of preterm deliveries

A → No. of abortion

L → No. of living children

Preterm → Before 37 Week

Early Term → 37wk - 38wk 6 Day's

Term → 39wk - 40wk 6 Day's

Late Term → 41wk - 41 wk 6 Day's

Post Term → > 42 wks

- EDD - Estimated date of pregnancy

→ Naegle's formula - Add 9 month + 7 day's

Symptom in Pregnancy

I - 1st Trimester symptom = 14 weeks

→ Earlier - Amenorrhea

→ Morning Sickness

→ ↑ Frequency of micturation

→ Lethargy

→ Breast discomfort d/to enlargement

II - 2nd Trimester

→ Quickening [Perception of fetal movements]

↳ MultiPara → ≥ 16 wks

↳ PrimiPara → ≥ 18 wks

III - 3rd Trimester

→ Lightening [Welcome Sign]

↳ MultiPara → At labour

↳ PrimiPara → 38wks

Sign's In Pregnancy

I - 1st Trimester

1. Goodel Sign → Softening Of Cervix

→ Earliest sign seen at 6weeks

2. Jacquimer Sign/ Chadwick Sign - 8wks

→ Bluish violet discoloration of cervix/ vagina due to venous congestion

3. Osiander Sign - 8 wks

→ Palpation Of Pulsation Of Uterine Vessel On Per Vaginal Examination

4. Piskacek's Sign - 8wks
→ Asymmetrical enlargement of uterus

5. Palmar Sign
→ Sense some tone D/T uterine contraction

6. Hegar's Sign/ Bimanual Sign
→ Abdominal & vaginal fingers seem to approximate each other

II - 2nd Trimester

→ Palpation of fetal parts ≥ 18wks

→ Auscultation of fetal heart sound ≥ 18wks
By stethoscope

III - 3rd Trimester

→ Lightening Sign

Investigation

→ Earliest - UPT [+ve after 1 day overdue]

→ Best - Blood [FIA > RIA] - +ve before 7- day overdue

→ Safest - USG →

	TVS	TAS
• Gestational Sac	4 wk	5 wk
• Yolk Sac	5 wk	6 wk
• Embryo	6 wk	7 wk

• Best clinical sign of pregnancy → Auscultation of fetal heart sound by stethoscope → 18wks

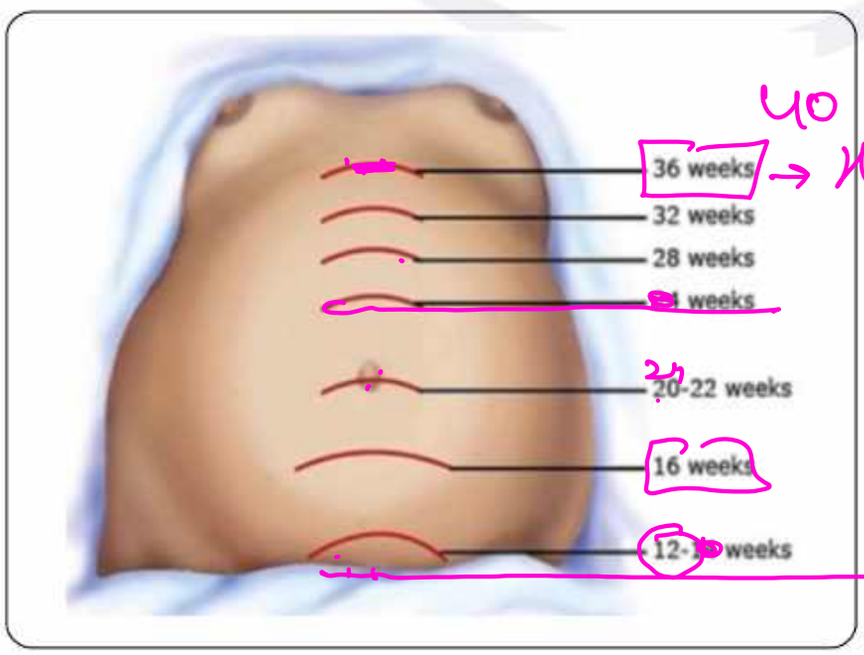
• Best Sign → Visualization of cardiac activity by USG

♦ Changes in Pregnancy

• Face

→ Blackish-Brown pigmentation on Cheeks, Nose, Fore Head d/to Melanin - ~~Melasma~~ = CHLASMA

- **Breast**
  - → ↑ Size
  - → Secondary Areola ↑
  - → Montgomery Tubercle
  - → Colostrum - Milk on squeezing breast
- **Abdomen**
  - → Linea Nigra = Brownish black line b/w xiphisternum & pubic symphysis
  - → Striae Gravidarum = Stretch marks
- **Vagina**
  - → Bluish Vagina
  - → pH - 3.5 [Acidic Vagina]
- **Wt.**
  - Total weight gain in pregnancy - 11-12kg
- **Fundal Height**
  - → Best clinical by - Size of abdomen
  - → Best overall by USG



♦ Ectopic Pregnancy

→ Implantation of fertilized ovum outside uterus

→ Mc R/F - H/O previous ectopic pregnancy

→ Mc Site - Ampulla

→ Mc site of rupture - Isthmus

Symptoms

→ Mc lower abdominal pain

Sign

→ P/V Exam - Empty uterus

→ Tubo ovarian mass +nt

→ Cervical motion tenderness

Ix -

→ β-HCG is doubled - Late developing intrauterine pregnancy

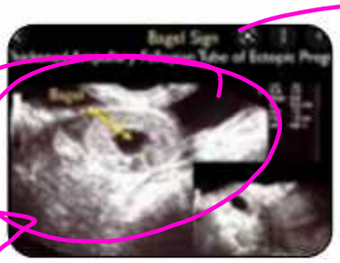
→ β-HCG ↓ - Abortion occurred

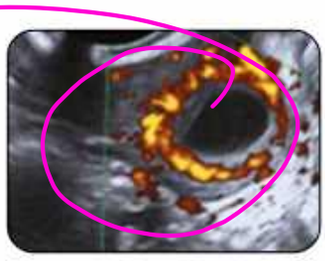
→ β-HCG ↑ But not double - Ectopic pregnancy

USG

→ Bagel Sign

→ Ring of fire on colour doppler





IXOC - TVS + S.B- HCG correlation

Gold Standard - Laparoscopy

- Management

1. Medical Mx

Indication - β-HCG - 1000-5000 IU/L

- Size Of Mass - <4cm

Doc - Methotrexate

If Jaundice - Actinomycin - D

2. Surgical Mx

Indication - No response to medical management

- β-HCG > 5000 IU/L

- Size of mass > 4cm

RxOC - Laparoscopy

→ If hemodynamically stable + Unruptured ectopic pregnancy
= Linear Salpingostomy

→ If hemodynamically unstable + Ruptured ectopic pregnancy
= Salpingectomy

♦ Molar Pregnancy

→ Abnormal proliferation & degeneration of chorionic tissue

R/F - Mc- ↓ Vitamin A, Folic Acid

- H/O molar pregnancy
- Smoking, OCP

Type - 1. Complete mole

2. Partial mole

1. Complete mole

→ Empty OVA + Sperm = 46XX

→ Uterus filled τ grape like vesicles

→ On USG - Snow storm appearance

2. Partial mole

→ 2 sperm + IOVA = 69XXY

→ On USG - Fetus & Ab. Placenta

Symptom of molar pregnancy

→ Mc- Irregular bleeding P/V

→ Amenorrhea

Ix

→ Gold standard - Histopath examination

→ β-HCG in complete mole → > 10^5 IU/L

→ β-HCG in partial mole → < 10^5 IU/L

Management

→ If Multiparous → Total Abdominal Hysterectomy

→ If Nulliparous → Suction Evacuation

• β-HCG Follow Up

Complication → Invasive Mole

→ Choriocarcinoma

♦ **Invasive Mole**

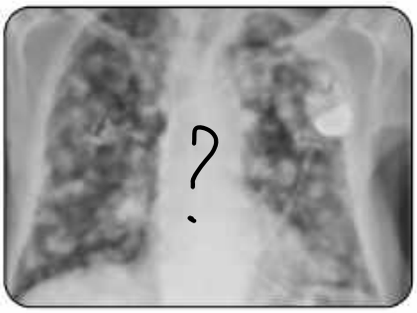
→ Invasion of molar preg. in myometrium

Mx - Hysterectomy

♦ **Choriocarcinoma**

→ Mc Symptom - Irregular bleeding per vagina

→ IxOC - β-HCG



→ Mc site of metastasis - Lung

Mx - For Low Risk - Methotrexate

For High Risk - Bagshaw Regime

E MAC O

E - Etopocide C - Cyclophosphamide

M - Methotrexate O - Oncovin

A - Actinomycind

♦ **APU - [Anti Partum Haemorrhage]**

→ Bleeding from genital tract after age of viability and before delivery of baby [28wks]

Mcc - Abruptio Placenta

Placenta Previa

Vasa Previa

Local Tear

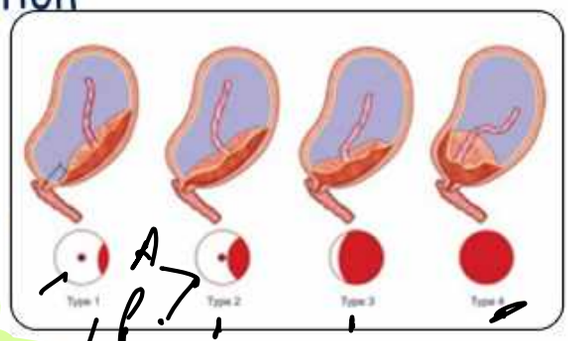
♦ **Placenta Previa**

→ If lower edge of placenta is within 2cm from internal Os

→ Mc R/F - H/O previous Placenta Previa

- H/O previous LSCS

→ Classification - Browne classification



→ Symptom - Bright red, painless, massive recurrent bleed

→ Sign - Abdomen & Uterus Soft & Non-Tender

→ IOC - TVS

PV Examination Is Mandatory

→ Mx Type I

Type II Ant. — NVD

Type II Post.
Type III
Type IV — C.S

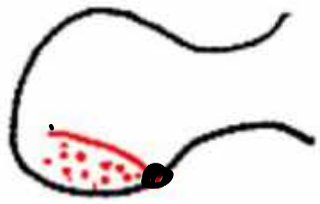
♦ **Abruptio Placenta**

Premature serration of placenta

Mc R/F - H/O previous abruption
- Polyhydroamnion
- Multifetal Pregnancy
- HTN

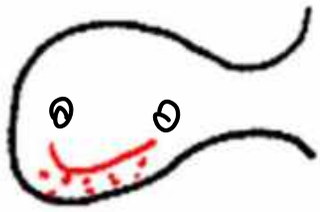
Type

1. Concealed



Lower edge of placenta is intact

2. Revealed



Lower edge of placenta is serrated

Symptoms → Abdominal pain

→ Continuous, dull red, painful P/v bleeding

Ix → USG + Clinical Correlation

Mx → Induction of labour if no DIC

→ If DIC is there - Rx FFP → Induction of labour

♦ Abortion

→ Death of fetus before age of viability [20 weeks]

Classification

A. USG based

1. Blighted Ovum

→ Only gestational sac, no embryo

2. Missed Abortion

→ Embryo without cardiac activity

B. Episode based

1. Spontaneous Abortion

→ Mc Cause - Aneuploidy >Trisomy

→ Mc Time - < 8 wks

→ Dx - Karyotyping

2. Recurrent Abortion

→ If H/O abortion ≥ 3

→ Mc Cause - Cervical Incompetence [2nd Trimester]

→ Dx - USG in pregnancy

↳Cervical length < 2cm

↳Internal OS > 2cm diameter

→ Mx - Mc Donald Stitches

C. Clinical presentation-based classification

1. Threatened Abortion

P/A - Fundal height = Period of amenorrhea

P/A - Internal O.S = Closed

USG - Live Fetus

Mx - Bed Rest

Progesterone

2. Inevitable Abortion = Fetus is in process of expulsion

P/A - Fundal HT ≤ POA

P/V - Int. OS = Open

Mx - Suction Evacuation

If gestational age > 12 add oxytocin

3. Incomplete Abortion

→ Abortion occurred but some products of conception are retained inside

P/A - Fundal HT ≤ POA

P/V - Int. OS - Open

Mx - Suction Evacuation

4. Missed Abortion

→ Abortion occurred but remained unnoticed

P/A Fundal HT < POA

P/V - Int. OS - Closed

Mx - Dilation & Evacuation

- MTP Act 2021
  - → Upto 20 week - 1 doctor
  - → B/w 20-24 week - 2 doctor
  - → MTP can be done upto 24 weeks
  - → For congenital malformation no age limit

- Method's
  1. First Trimester
     - → Mc - Medical Abortion [Upto 7 week's]
     - → Best Suction Evacuation

     - → Medical Abortion

       Day 1 - Tab. Mifepristone

       After 48 Hrs - Tab. Mistoprostol [PGE-1]

       ↓ if fail's

       → Suction & Evacuation

       ↳Karman Cannula

2. IInd Trimester

Mc & Best → Prostaglandins

Doc - Dinoprostone 20mg gel

Carboprost - 250mg inj.

♦ Pregnancy Induced Hypertension

1. Gestational Hypertension

→ ↑ BP after 20 was of pregnancy which become normal within 12wks of delivery

Doc - Labetalol

2. Pre-Eclampsia = ↑ BP + HTN

Mild Pre-Eclampsia	Severe Preeclampsia
SBP 140 - 160	> 160
DBP 90-110	> 110
Protein Urea < 5g/ 24 hrs	> 5g/ 24 hrs
Dipstick Test +1 - +3	> +3
	• End organ damage
	• ↓ Vision + Headache
Rx - Monitoring till 37wk	• IUGR
→ Induction of labour	• HELLP Syndrome

Rx - Doc - Labetalol

→ Monitor till 34 wk →
Induction or labour

- HELLP Syndrome

Δ → TENSSE criteria

Platelets < 1 lakh
LDH > 600 IU
Liver enzymes > 70 IU

Mx → Termination of Pregnancy

3. Eclampsia

Severe Pre-Eclampsia + Convulsion [Due to cerebellar hypoxia]

Mx - 1st-step - Maintain airway

↓

Doc to prevent & Rx eclampsia $MgSO_4$

↓

Labetalol - For Preeclampsia

↓

Immediate induction of labour irrespective of gestational age

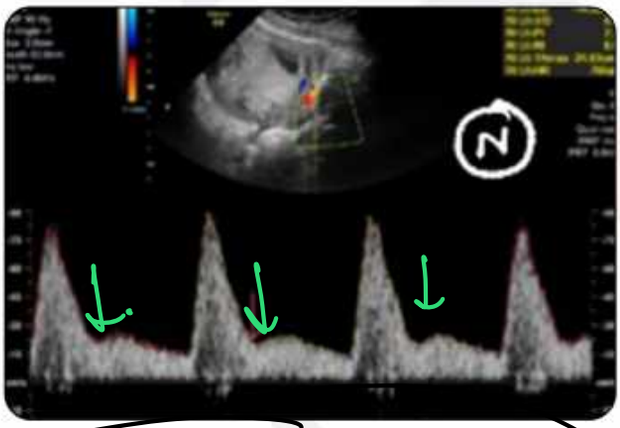
→ Doc for HTN emergency → I.V Labetalol

→ Doc for chronic HTN → Labetalol

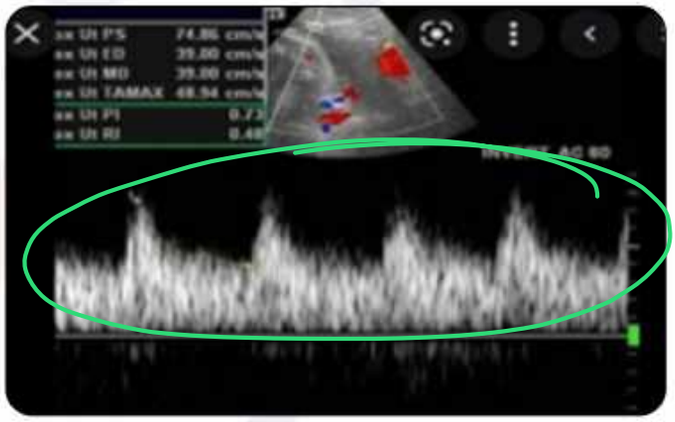
→ Doc for refractory HTN → Sodium Nitroprusside

→ Doc for status eclampticus → Thiopentone Sodium

→ Predictor Of PIH → Uterine Artery Colour Doppler



< 22 wks Diastolic Notch +nt.



> 22 wks - Diastolic Notch Disappear

↳ Indicate HTN.

♦ **RH Incompatibility**

→ If fetus is RhD positive

&

→ Mother is RhD negative

↓

→ IgM is formed & IgM is big so unable to cross placenta hence 1st preg. is safe

↓

→ In next pregnancy if fetus is RhD +ve

↓

→ IgM is converted into IgG & IgG can cross placenta

↓

→ Fetal Hemolysis

↓

→ Hypoxia

↳ Complication

1. Liver Damaged
2. Hepato Splenomegaly
3. Polyhydroamnios - Golden Yellow Baby
4. Hydrops Fetal is [> 2 = Δ Hydrop's Fetalis]

↳ Skin Edema
↳ Ascites
↳ Hydrothorax
↳ Pericardial Eff.

→ Earliest evidence of Hydrops Fetalis → Scalp Edema [Buddha Sign]

Diagnosis - Coomb Test

→ If ICT -ve → Give anti-D 300 µg to mother at 28 weeks of pregnancy

→ If ICT +ve → G.A >= 34 week → Induction of labour

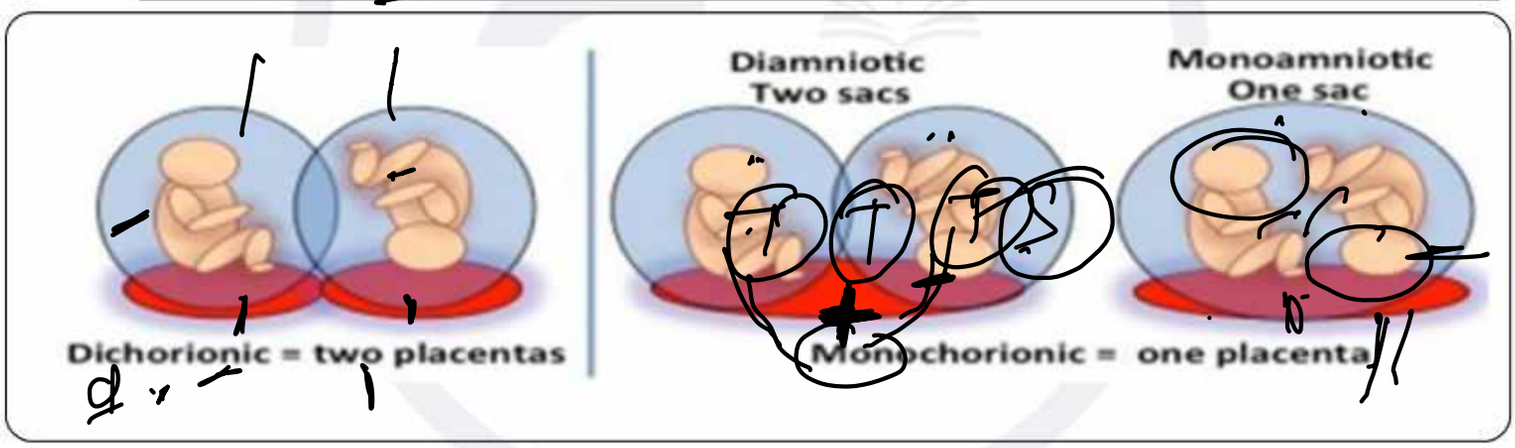
II Direct Coomb Test = For new born

→ If DCT -ve → 2nd dose of anti-D within 72-hour upto 28 days

♦ Multi Fetal Pregnancy

Mc - Twin

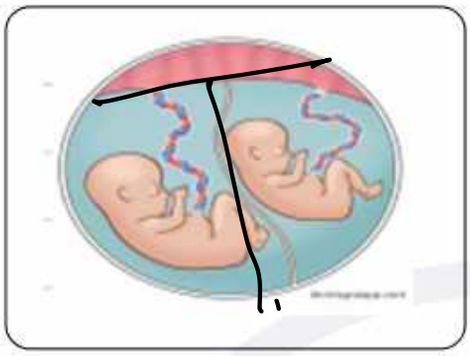
Monozygotic	Dizygotic
→ 1 Zygoto divided into 2	→ From 2 diff. Zygote
→ Identical Twins	→ Non-identical twin
→ Blood group & sex same	→ Can be same can be different
→ Can be monochromic or dichorionic	→ Always Dichorionic



- Type Of Monozygotic Twins

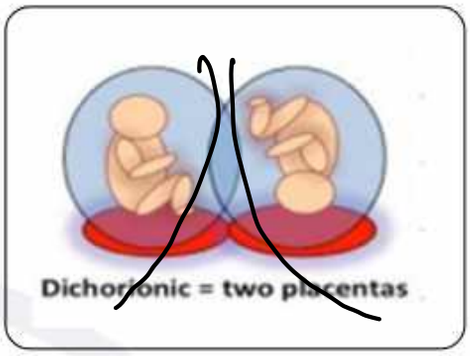
→ Division within 3 days → Dichorion - Diamnion
→ 4 - 8 days → Monochrionic - Diamnion
→ 9 - 12 days → Monochromic - Monoamnion
→ After 12 days → Conjoint Twins
↳ Paraphagus > Thoracophagus

Monochorionic



→ T or inverter T sign

Dichorionic



→ Twin peak sign or λ sign

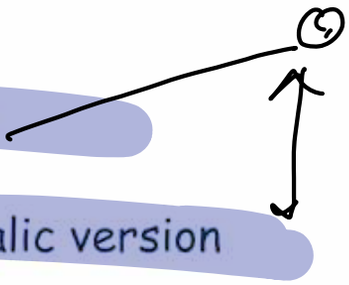
- Complication of Monochromic Monoamniotic
  - → Cord Entanglement
- Complication of Monochromic Diamniotic
  - → Twin - Twin Transfusion Syndrome
- Delivery

1. Mc Presentation - Both Cephalic
   - Mx → Vaginal Delivery
   - Rule → If 1st baby is non-cephalic - DO - CS
2. 1st Cephalic & 2nd Breech
   - 1st Cephalic → Vaginal delivery
   - 2nd Breech → Assisted breech delivery

3. 1st Cephalic & 2nd Transverse

1st Cephalic → Vaginal Delivery

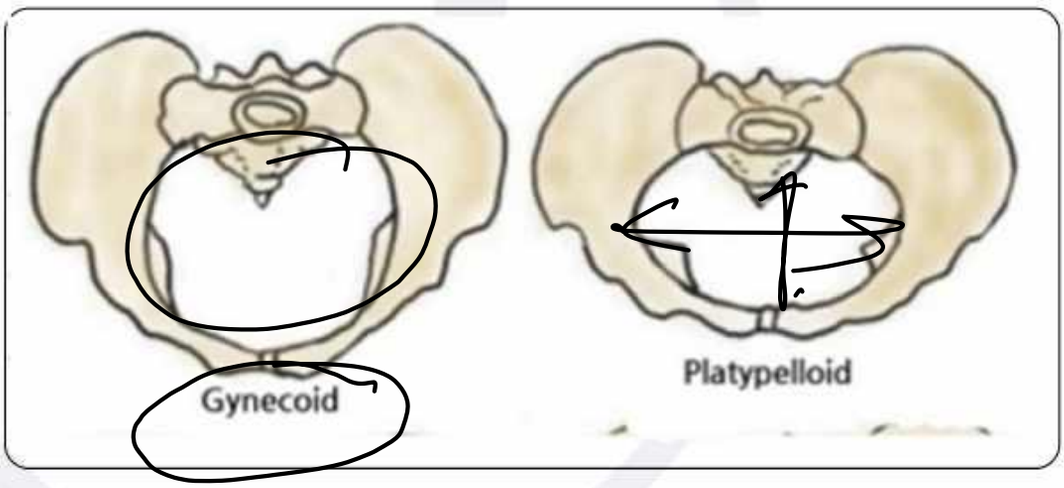
2nd Transverse → Internal podalic version



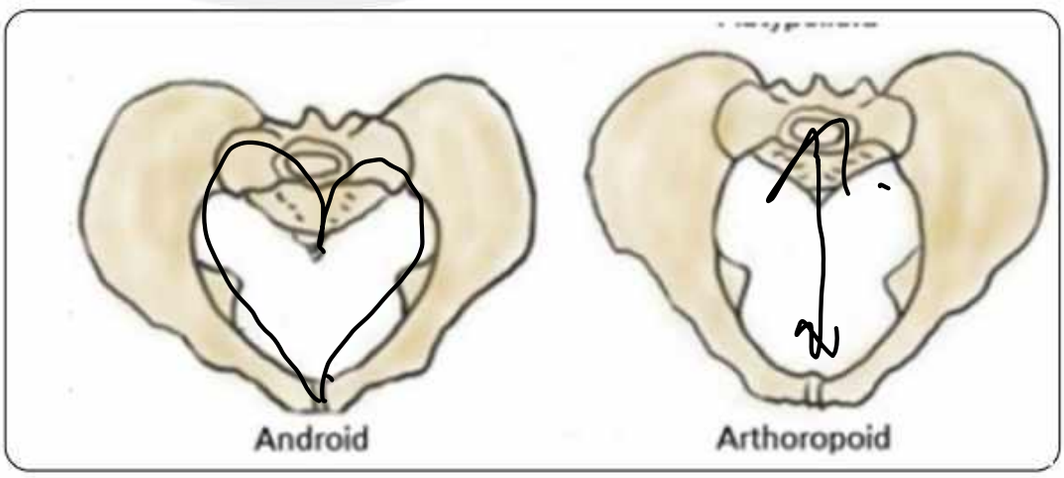
♦ Types of Pelvis

↳ Caldwell Moloy Classification

Round Inlet



Heart Shape



→ Contracted Pelvis → Any diameter < 1cm

1. Congenitally Contracted Pelvis

A. Naegle's Pelvis - One ALA of sacrum -nt

B. Robert's Pelvis - Both ALA of sacrum -nt

2. Acquired Contracted Pelvis

↓ Vit = D

A. In children ↑ severe rickets - Bean shape pelvis

B. Adult ↑ Osteomalacia - Triradiate Pelvis

Dx - MRI

Mx - C-Section

♦ **CPD- Cephalo Pelvic Disproportion**

→ Size of fetal head & pelvic are out of proportion

→ Dx - MRI

→ Rx - Mild CPD -Trial of labour

- Severe - C-Section

♦ **Fetal Skull diameter**

I. Transverse diameter

→ Bimastoid -7.5cm

→ Bitemporal - 8cm

→ Biparietal - 9.5cm [Mc Transverse engaging diameter]

II. Longitudinal diameter

→ Suboccipito Bregmatic - 9.5cm (Mc)

→ Mento vertical - 14cm (Largest)

• Presenting Part

→ Part of presentation which is above internal OS

Example - Vertex, Brow, Face

• Denominator

→ Bony prominence on presenting part

Example - Vertex- Occiput

- Face - Mentum (chin)

♦ **Lie**

Relation b/w maternal & fetal spine

♦ **Station**

→ Relation b/w presenting part & ischial spine

→ At Ischial spine → 0

→ Above Ischial spine → -1

→ Below Ischial spine → +1

♦ **Position**

→ Relationship b/w denominator & pelvis

→ Mc - LOT > LOA

♦ Labour

4 Stages

→ Stage I → From true uterine contraction to complete cervical dilatation

→ Stage II → From complete cervical dilatation to delivery of baby

→ Stage III → Placental expulsion

→ Stage IV → 1hr observation after placental expulsion

♦ Stage 1 -

Latent Phase

→ Upto 5cm dilation

- Duration
  - ↳ Primi - Upto 12 hrs
  - ↳ Multi - Upto 8 hrs
- Delayed
  - ↳ Primi - ≥ 20 hrs
  - ↳ Multi - ≥ 14 hrs
- Mx - Therapeutic rest

Active Phase

→ 6-10cm dilation

- Arrest of active phase

→ No cervical dilation for 4 hrs despite good uterine contraction

- Mx - C- section

- Friedmann - Partogram
  - → Best method to assess progress of labour
  - → If alert line is crossed → Prolonged labour
  - → If action line is crossed → C- Section

→ Time gap b/w alert & action line → 4 hrs

♦ **Stage 2-**

→ Engagement - Max. transverse engaging diameter enter Pelvic Brim

↓

→ Descent -

↓

→ Flexion

↓

→ Inter Rotation - Occur at Ischial spine

↓

→ Crowning - Permanent visualisation of scalp hair even when uterine cont. is over

↓

→ Extension

↓

→ Restitution - Untwisting of neck

↓

→ External Rotation - Passive movement of head outside due to internal rotation of shoulder inside 45°C

Mx Of Delay
→ C- Section
Primi - ≥ 3 hrs
Multi - ≥ 2 hrs

♦ **Stage 3**

→ Keep baby tray below maternal level

→ Doc - Oxytocin

→ Look for the sign of placental serration

- Abdominal - Firm, globular, cricket ball feel uterus
- Vaginal - Can feel placenta in vagina, gush of blood out of vagina

Mx - Inj. oxytocin if delayed ≥30 min.

♦ **Stage 4**

→ 1hr observation after placental expulsion

→ Physiological chills seen

♦ **Pressure**

→ Stage 1 → 40-50 mmHg

→ Stage 2/3 → 100-120 mmHg

♦ **Mc-Method of placental separation**

→ Stage 1 → 40-50 mmHg

→ Stage 2/3 → 100-120 mmHg

♦ **Uterine Inversion**

Mcc - Mismanaged 3rd stage of labour

Stages of inversion

1° - Dimpling of uterus

2° - Upto Int. Os

3° - Upto Ext Os

4° - Whole uterus turned inside out

Mx - Immediate Resuscitation

↓

Manual Reposition [Johnson Method]

↓

O "Sullivan" Hydrostatic method

↓

Surgery - Huntington method

♦ **Preterm Labour**

→ If labour before 37 wks

R/F → H/O previous preterm labour

→ Polyhydroamnion

→ Macrosomia

→ Multifetal pregnancy

Mcc → Idiopathic > Infection

↳ Gardnerella -Mc

↳ Chlamydia

↳ Listeria

Diagnosis - Cervical Length - <2cm

Int. Os Diameter - >3cm

Fetal Fibronectin - >50 ng/ ml

Mx - Steroid - Dexamethasone

Tocolytic - Nifedipine

In heart disease - Atosiban

MgSO4 - If G.A <28 wks

If G.A >34 wks - Wait & watch

♦ **Prom-Premature rupture of membrane**

→ Membrane rupture before onset of labour

→ Mcc - Infection, Polyhydroamnion, Macrosomia

→ Diagnosis → Litmus paper test - Turn Red → Blue

→ Ferning pattern seen on microscopy

→ Fetal Fibronectin

Best → USG - Oligohydroamnion

Mx - If Ga < 34 wks

M - $MgSO_4$ If GA <28wks

A - Antibiotic

S - Steroid

T - Tocalytics

If GA > 34 wks

Antibiotics

Induction of labour

♦ Post Term Pregnancy

→ If > 42 wks

→ Mcc - Mistaken dates → Ask to review Hx

→ Complication - Calcification in placenta

Mx - Induction of labour

♦ Induction of labour

Medical - Doc PGE2 - Dinoprostone

Surgical - Sweeping/ stripping of membrane

♦ Augmentation Of Labour

Medical - Oxytocin

Surgical - Artificial rupture of membrane by Kocher's Forceps

♦ Obstructed Labour

→ Prolongation of labour despite of good uterine contraction

C/F → Bandl's Ring - Pathological ring at upper segment

Pt Exhausted

Vagina Hot & Dry

Haematuria

Mx - Immediate C- section

♦ Post Term Pregnancy

→ Constriction ring due to injudicious use of oxytocin

→ Physiological seen only vaginally

Mx - May resolve spontaneously → C.S

♦ PPH - Postpartum Haemorrhage

Types

I. Primary PPH → Within 24 hrs Of Delivery

→ Mcc - Uterine Atony

II. Secondary PPH → After 24hrs & before 12 wks of delivery

→ Mcc - Retained placenta

Mx of PPH

→ Injection tranesmic acid 1gm + Ressucitation

↓

→ Doc - Oxytocin

↓

→ Oxytocin + Uterine massage

↓

→ Best Drug - Carboprost or Hembate

↓

→ BiManual uterine compression

→ External aorta compression

→ Military anti shock garments

→ Bakri balloon catheter

→ Condom catheter

If No Response

Multi-Para

Hysterectomy

Primi-Para

→ Compression B. Lynch Sutures

→ Uterine Artery Ligation B/L

→ Ovarian Artery Ligation B/L

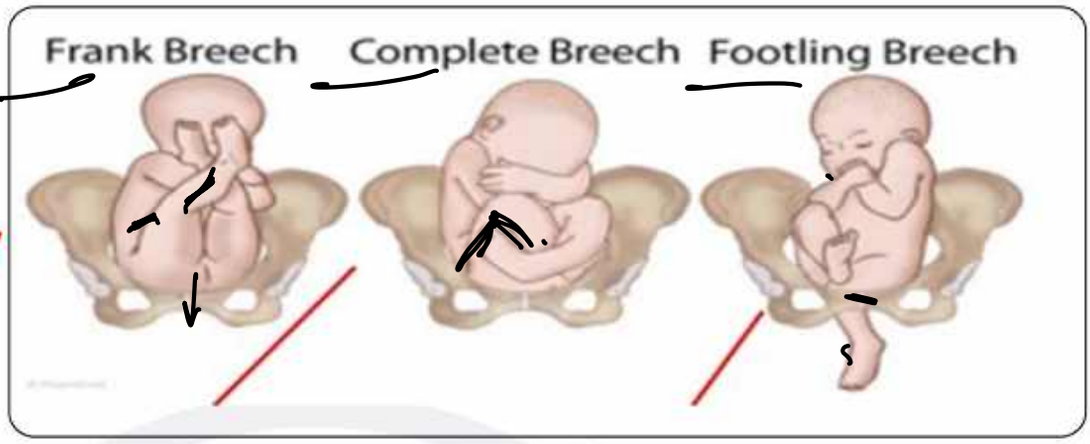
Hysterectomy

↳ Umbrella Pack

↳ Parachute Pack

♦ Breech

Mcc - Prematurity



Frank Breech	Complete Breech	Footling Breech
→ Thigh flexed & knee extended	→ Than flexed & knee flexed	→ Thigh & knee extended
→ PV - Buttocks genitalia	→ PV - Buttocks, genitalia, feet	→ PV - Feet

Mx of Breech

1. For extended leg - Pinnard Maneuver
2. For extended arm - Loveset Maneuver
3. To Deliver After Coming Head
   - → Developed countries - Piper Forceps
   - → Other method - Burn Martial method
     - ↳ If after coming head is entrapped

↓

Preterm → Dushrren incision on cervix - 10° clock - 2% clock

Term → Zavenelli Maneuver - Push fetus inside & do C- Section

→ If Stargazer Breech → Hyper extended head

Mx – C-section

♦ Transverse

Mcc - Prematurity > Multiparity

Mx - External cephalic version at 36 week's under USG - Guidance

If she is in labour - C- Section

♦ Face

↳If Mento anterior - Vaginal delivery

↳If Mento posterior – C-section

♦ Instrumental Delivery

A - Forceps Delivery

Pre-requisites → Station ≥ +2
Complete dilation
Membrane ruptured
No Contracted pelvis

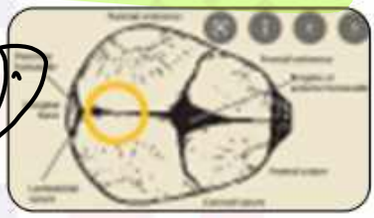
Traction Force → PRIMI 18-20kg
Multi 13kg

Max Pull - 3 if fail's - C.S

Indication - Maternal & fetal distress

B Vaccum delivery

6cm



Instrument - Silastic Cup

Applied On - Applied on sagittal suture 3cm ant. to posterior fontanelle

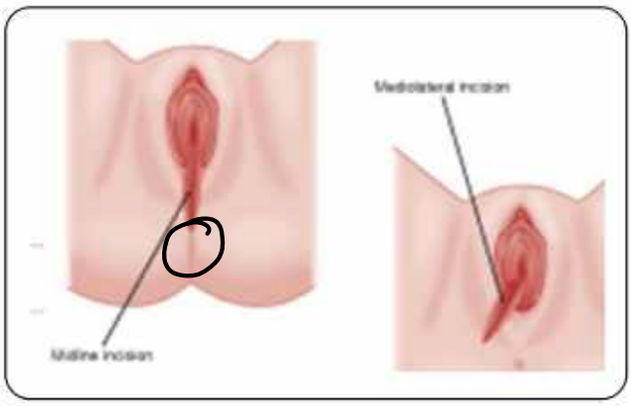
Suction Pressure - 0.2 - 0.8kg/ cm2

Max Pull - 3 if fail's - C-section

Advantage - Only 6cm dilation is enough

Disadvantage - Can't use in fetal distress
More fetal injury
Cephalohaematoma

♦ Episiotomy



Incision on post. vaginal wall to ↑ vulval outlet

Indication - Primi - Rigid perineum
Instrumental delivery
Shoulder dystocia

Types - Median
Mediolateral

♦ C-Section

Indication - Contracted Pelvis
CPD etc.

Types Of Incision

1. Abdominal Incision
   - Mc -Low Transverse - Pfannenstiel Incision
2. Uterine Incision
   - Mc - Low traverse - Kerr Incision

♦ Puerperium

- 6wks period after delivery in which all organ's return back to normalcy

↳ Weight of Uterus

- Immediately after delivery - 1000 grm

- After 1 week - 500grm
- After 2 week - 300grm
- After 3 week - 100grm
- After 6 week - 60-80 grm

↳ Post-Partum Discharge [Lochia]

- Lochia Rubra - RBC - Red [1-4 day's]
- Lochia Serosa - WBC - Yellow [5-9 day's]
- Lochia Alba - Decidua - White [10-15 day's]

↳ Post-Partum blue

Due to withdraw of estrogen/ progesterone there is nervousness, anxiety in 1st 10 day's

Mx - Reassurance

↳ Post-Partum depression

- Seen after 4-6 month 2-3

Mx - Fluoxetine

OR Box.

Q 1.

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